Frequentist Estimation of the Epistemological Paradigm–Skill Pedagogy Index (EPSPI):

An Eleven-Method Comparative Framework with
Three-Layer Architecture and Episode-Level Evaluation

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1 Introduction

This paper makes the following contributions:

(i) A formal probabilistic definition of EPSPI grounded in Laplace-smoothed conditional probability estimation.

(ii) A three-layer computational architecture that preserves within-episode information (mutual information = 1.24 bits, retaining 60.5% of within-episode structure) while enabling episode-level evaluation.

(iii) Eleven frequentist estimation methods organised along three methodological axes: direct estimation, representation learning, and predictive modelling.

(iv) A comprehensive evaluation framework with six metrics including cross-validated Pearson r, calibrated explained variance, AIC, and BIC.

(v) A leverage analysis identifying the structural drivers of the performance ceiling.

2 Data, Notation and EPSPI Probabilistic Formulation

2.1 Corpus and Variables

The corpus consists of $N = 19,181$ classroom utterance lines and teacher excerpts drawn from 49 episodes: Episodes 1–30 are Science lessons (grades 6–12), Episodes 31–48 are Mathematics lessons (grades 6–12), and Episode 49 is a vignette containing teacher excerpts from a grade-7 science lesson. Learning theories are referred to as Educational Practices (EP) in this study. Each line n is annotated with binary EP presence indicators and binary skill presence indicators:

- $X_{ni} \in \{0, 1\}$: EP i is active in line n, $i \in \{1, \ldots, P\}$ (Behaviourism, Cognitivism, Constructivism; $P = 3$).
- $Y_{nj} \in \{0, 1\}$: Skill j is present in line n, $j \in \{1, \ldots, K\}$ (Problem Solving, Creativity, Collaborative Learning, Critical Thinking; $K = 4$).
- $e_n \in \{1, \ldots, 49\}$: Episode of line n.
- $n_e = \{n : e_n = e\}$: The set of all lines n whose episode label equals e.

1	Global encoding	Speaker	Transcript	Time		Knowledge				Cognitive Process						LT			21st CS		
2	Preassessment			stamp	FK	CK	PK	MK	Rem	Und	App	Ana	Eva	Cre	Beh	Cog	Con	PS	C	WC	CT
88			no group wise when one group is there other group	00:05:03	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	3	0
89			will not com you can sit anywhere but need not come here	00:05:06	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
90			this line is known as	00:05:12	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
91		speaker2	stem	00:05:12	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
92		Speaker1	no what it will look like what has leader done	00:05:16	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	3	0
93			first leader you tell me	00:05:17	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	3	0
94			what is this and it which type of venation	00:05:20	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
95		speaker2	reticulation	00:05:23	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
96		Speaker1	and this one	00:05:25	0	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
97		speaker2	sir this is parallel	00:05:26	0	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
98			sir this is parallel	00:05:30	0	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
99		Speaker1	learn again	00:05:22	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	0
100			learn again here come next group	00:05:33	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	0
101			now tell me which venation	00:05:36	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
102		speaker2	sir parallel venation	00:05:39	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
103		Speaker1	why parallel venation	00:05:41	1	0	0	0	0	2	0	0	0	0	1	2	0	0	0	0	0
104		speaker2	sir because this thing is going in	00:05:43	0	0	0	0	0	2	0	0	0	0	1	2	0	0	0	0	0
105		Speaker1	what is this thing what is this thing	00:05:44	0	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
106		speaker2	veins	00:05:45	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
107		Speaker1	where is the vein sophia?	00:05:46	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
108			where is the vein ?	00:05:48	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
109			see can you see the vein?	00:05:50	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
110			can you see the vein?	00:05:53	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
111		speaker2	sir this	00:05:55	0	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
112		Speaker1	uh, this is?	00:05:57	0	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
113			and this thicker line is known as	00:05:59	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
114		speaker2	midrib	00:06:00	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
115		Speaker1	thicker line is known as	00:06:01	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
116		speaker2	midrib	00:06:04	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0
117		Speaker1	another bring bigger leaf bigger leaf	00:06:10	1	0	0	0	1	0	0	0	0	0	1	2	0	0	0	0	0

Figure 1: Class-wise vs variables for Science subjects plots

2.2 Marginal, Joint, and Conditional Probabilities

For a randomly selected line n, the marginal probability of EP_i being active and the marginal probability of skill S_j being present are:

$$P(EP_i) = \frac{\sum_{n=1}^{N} X_{ni}}{N}, \quad P(S_j) = \frac{\sum_{n=1}^{N} Y_{nj}}{N} \tag{1}$$

The joint probability of EP_i and S_j co-occurring in the same line is:

$$P(EP_i \cap S_j) = \frac{\sum_{n=1}^{N} X_{ni} Y_{nj}}{N} \tag{2}$$

The conditional probability of skill S_j given that EP_i is active follows directly:

$$P(S_j|EP_i) = \frac{P(EP_i \cap S_j)}{P(EP_i)} = \frac{\sum_{n=1}^{N} X_{ni} Y_{nj}}{\sum_{n=1}^{N} X_{ni}} \tag{3}$$

Laplace smoothing: Since constructivism is absent from many individual episodes ($\sum_{\{n:e_n=e\}} X_{n,\mathrm{Con}} = 0$ for most e), the denominator of Equation (3) is zero. To resolve this, we use Laplace (additive) smoothing with parameters $\alpha = 1$ and $K = 4$, adding a small pseudo-count to both numerator and denominator. The smoothed estimate, denoted θ_{ij}, is:

$$P(S_j|EP_i) = \theta_{ij} = \frac{\sum_{n=1}^{N} X_{ni} Y_{nj} + \alpha}{\sum_{n=1}^{N} X_{ni} + K\alpha} \tag{4}$$

This is the global conditional probability Θ, pooling all N lines. The resulting 3×4 matrix $\Theta = [\theta_{ij}]$ is the core quantity of the EPSPI framework; all eleven estimation methods are either direct estimators of θ_{ij} or functions thereof. Observed values are reported in Table 1.

Table 1: Laplace-Smoothed Θ

	S1: PS	S2: C	S3: CL	S4: CT
EP1: Beh	0.3239	0.0071	0.0360	0.0982
EP2: Cog	0.2138	0.0069	0.0094	0.3591
EP3: Con	0.8061	0.0418	0.8783	0.3118

Constructivism exhibits the highest conditional probabilities for Problem Solving and Collaborative Learning by a wide margin, reflecting a near-perfect association between Constructivist pedagogy and collaborative skill deployment ($\theta_{\text{Cog,CL}} = 0.0094$ vs. $\theta_{\text{Con,CL}} = 0.8783$). This extreme contrast causes quasi-separation in maximum-likelihood methods. To quantify estimation uncertainty, 95% confidence intervals for $\hat{\theta}_{ij}$ were computed using the Wilson score interval for binomial proportions [31]. The Wilson interval provides better coverage than the normal approximation, particularly for small probabilities.

Let $n_i = \sum_n X_{ni}$ and $n_{ij} = \sum_n X_{ni} Y_{nj}$. The Wilson interval for θ_{ij} with confidence level $1 - \alpha$ is given by

$$CI_{ij} = \frac{\hat{\theta}_{ij} + \frac{z^2}{2n_i} \pm z\sqrt{\frac{\hat{\theta}_{ij}(1-\hat{\theta}_{ij})}{n_i} + \frac{z^2}{4n_i^2}}}{1 + \frac{z^2}{n_i}}, \tag{5}$$

where $z = 1.96$ for a 95% confidence level. Confidence intervals for Behaviourism and

Table 2: Global conditional probability matrix Θ with 95% Wilson confidence intervals. Entries are $P(\text{Skill } j \mid \text{EP}_i)$. (PS: Prob. Solving, Cr: Creativity, CL: Collab. Learning, CT: Critical Thinking)

	PS	Cr	CL	CT
Behaviourism	0.324 [0.314, 0.334]	0.007 [0.005, 0.009]	0.036 [0.032, 0.040]	0.098 [0.092, 0.105]
Cognitivism	0.214 [0.207, 0.221]	0.007 [0.006, 0.008]	0.009 [0.008, 0.011]	0.359 [0.351, 0.367]
Constructivism	0.806 [0.763, 0.857]	0.042 [0.021, 0.070]	0.878 [0.844, 0.921]	0.312 [0.259, 0.372]

Cognitivism are relatively narrow due to large sample sizes ($n_i = 8601$ and $n_i = 14422$), while Constructivism exhibits wider intervals ($n_i = 259$), reflecting greater estimation uncertainty.

2.3 Episode-Conditional Probabilities

The global θ_{ij} pools information across all 49 episodes. To capture local pedagogical context within a specific episode, we define the episode-conditional probability:

$$\theta_{ij}^{e} = \frac{\sum_{\{n:e_n=e\}} X_{ni}Y_{nj} + \alpha}{\sum_{\{n:e_n=e\}} X_{ni} + K\alpha} \tag{6}$$

When EP_i is entirely absent from episode e (denominator $= K\alpha = 4$ before smoothing), the formula returns $\theta_{ij} = \frac{1}{K} = 0.25$, the maximum-entropy prior over four equally probable skills.

3 Information-Theoretic Justification

To justify the requirement for a line-level analysis, we employ an information-theoretic framework to quantify the "structural intensity" of the pedagogical patterns and determine the information loss associated with data aggregation.

3.1 Entropy and Mutual Information

We define the configuration pattern of a line n as a 7-bit binary string $z_n \in \{0,1\}^7$, representing the simultaneous state of $P = 3$ Educational Practices and $K = 4$ Skills. The global uncertainty of these patterns is measured by the **Global Entropy** $H(z)$:

$$H(z) = -\sum_{z\in\mathcal{Z}} p(z)\log_2 p(z) \approx 3.140 \text{ bits} \tag{7}$$

To determine the influence of the episode context, we calculate the **Conditional Entropy** $H(z|e)$, which measures the average remaining uncertainty of a pattern given its episode label e:

$$H(z|e) = \sum_{e=1}^{E} \frac{n_e}{N}\left[-\sum_{z\in\mathcal{Z}} p(z|e)\log_2 p(z|e)\right] \approx 1.901 \text{ bits} \tag{8}$$

The **Mutual Information** $I(z,e)$ quantifies the information shared between pedagogical patterns and their episode context:

$$I(z,e) = H(z) - H(z|e) \approx 1.239 \text{ bits} \tag{9}$$

The **Relative Information** ($I/H \approx 39.5\%$) reveals that while episode-level context is significant, 60.5% of the information variance resides at the line-level. This high "intra-episode" variance necessitates the *Three-Layer Architecture* to prevent the loss of

fine-grained pedagogical data.

3.2 Divergence Metrics: Structural Intensity

To identify specific episodes that deviate from the global pedagogical norm, we define two divergence measures comparing the episode distribution P to the global distribution Q.

3.2.1 Normalized KL Divergence

We use the Kullback–Leibler (KL) divergence to measure the "surprise" of an episode's distribution relative to the corpus, normalized by global entropy to define a **Structural Intensity Index**:

$$D_{KL}(P \parallel Q) = \sum_{z \in \mathcal{Z}} P(z) \log_2 \left(\frac{P(z)}{Q(z)} \right), \quad \text{Norm KL}_e = \frac{D_{KL}(P \parallel Q)}{H(z)} \tag{10}$$

3.2.2 Jensen–Shannon Divergence (JSD)

To provide a symmetric and bounded distance metric ($0 \leq JSD \leq 1$), we utilize the Jensen–Shannon Divergence, which compares both distributions to an average $M = \frac{1}{2}(P + Q)$:

$$JSD(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M) \tag{11}$$

3.3 Permutation Test for Statistical Significance

To ensure the observed $I(z, e)$ is not a result of the large number of episodes ($E = 49$), a permutation test was conducted ($2,000$ iterations). Under the null hypothesis H_0 (no relationship between pattern and episode), the episode labels were randomly shuffled.

- **Observed MI:** 1.239 bits
- **Null Distribution:** $\mu = 0.053, \sigma = 0.0015$
- **p-value:** < 0.0005

The observed MI is significantly higher (> 700 standard deviations) than the null mean, confirming that the pedagogical structures are highly episode-dependent and non-random.

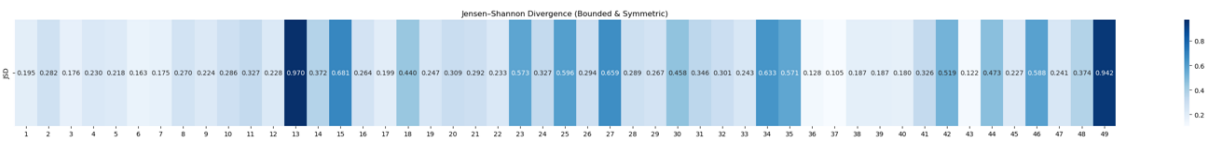

Figure 2: Class-wise vs variables for Science subjects plots

Table 3: Entropy and Information Results

Metric	Value
Global Entropy $H(z)$	3.140 bits
Conditional Entropy $H(z\|e)$	1.901 bits
Mutual Information $I(z, e)$	1.239 bits
Relative Information Explained	39.47%

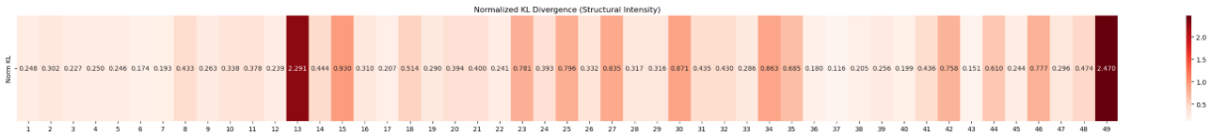

Figure 3: Class-wise vs variables for Science subjects plots

3.4 Three-Layer EPSPI Architecture

The EPSPI framework is constructed in three layers to preserve within-episode information before aggregation.

Layer 1: Line-level EPSPI

$$\text{EPSPI}_n = f(X_{n\cdot}, \Theta, \hat{\theta}^{e_n}), \quad n = 1, \ldots, N \tag{12}$$

Layer 2: Episode aggregation

$$\text{EPSPI}_e = \frac{1}{n_e} \sum_{\{n:e_n=e\}} \text{EPSPI}_n \tag{13}$$

Layer 3: Validation target The observed skill rate for episode e, computed independently from all EP and EPSPI quantities, is:

$$\bar{Y}_e = \frac{1}{n_e} \sum_{\{n:e_n=e\}} \left(\frac{1}{K} \sum_{j=1}^{K} Y_{nj} \right) \tag{14}$$

The primary validation criterion is the cross-validated correlation CV $r(\text{EPSPI}_e, \bar{Y}_e)$. The decomposition $\bar{Y}_e = \text{EPSPI}_e + \varepsilon_e$ defines the non-EP residual ε_e.

4 Evaluation Framework

4.1 Episode-Level Metrics

All primary metrics are computed at the episode level ($E = 49$) to respect the hierarchical structure of the data.

Train Pearson r.

$$r_{\text{train}} = \operatorname{corr}\big(\text{EPSPI}_e,\, \bar{Y}_e\big), \quad e = 1, \ldots, 49.$$

Calibrated Explained Variance (Exp.Var(cal)). An OLS calibration $\hat{Y}_e = a + b \cdot \text{EPSPI}_e$ is fitted; explained variance is then:

$$\text{Exp.Var(cal)} = 1 - \frac{\operatorname{Var}(\bar{Y}_e - \hat{Y}_e^{\text{cal}})}{\operatorname{Var}(\bar{Y}_e)}.$$

Calibration corrects for scale differences between EPSPI and $\bar{Y}_e$.

Raw Explained Variance (Exp.Var(raw)).

$$\text{Exp.Var(raw)} = 1 - \frac{\operatorname{Var}(\bar{Y}_e - \text{EPSPI}_e)}{\operatorname{Var}(\bar{Y}_e)}.$$

Negative values indicate the raw EPSPI scale is offset from $\bar{Y}_e$; the gap Exp.Var(cal) − Exp.Var(raw) quantifies scale bias per method.

Cross-Validated Pearson r. Five-fold CV with GroupKFold splits at the episode level (no line from a test episode appears in the training fold):

$$\text{CV}\, r = \frac{1}{5} \sum_{k=1}^{5} \operatorname{corr}\big(\text{EPSPI}_e^{(k)},\, \bar{Y}_e\big), \quad e \in \text{test fold } k.$$

Cross-Validated MSE.

$$\text{CV MSE} = \frac{1}{5} \sum_{k=1}^{5} \frac{1}{|F_k|} \sum_{e \in F_k} \big(\text{EPSPI}_e - \bar{Y}_e\big)^2.$$

4.2 Line-Level Metrics

AIC and BIC. For each method, a fixed external logistic model is fitted at the line level:

$$\operatorname{logit}\big(P(Y_{nj} = 1)\big) = \beta_{0j} + \beta_{1j} \cdot \text{EPSPI}_n, \quad j = 1, \ldots, 4.$$

AIC and BIC are computed from the log-likelihood of this model, summed across all four skills:

$$\text{AIC} = \sum_{j=1}^{4} \big(2k_j - 2\ell_j\big), \quad \text{BIC} = \sum_{j=1}^{4} \big(k_j \ln N - 2\ell_j\big),$$

where $k_j = 2$ (intercept + slope) and ℓ_j is the maximised log-likelihood. Because k_j and N are identical across all methods, AIC/BIC differences reflect log-likelihood differences exclusively.

Remark 1. AIC and BIC are meaningful here even for non-parametric methods (RF, SVM, NMF) because the likelihood is computed from the fixed external logistic model, not from the EPSPI method itself. This provides a consistent line-level discriminative comparison across all eleven methods.

4.3 Rationale and Worked Examples for Each Metric

The six metrics collectively address three distinct questions about an EPSPI method: (1) does it *rank* episodes correctly? (2) does it explain episode-level *variance* in skill deployment? and (3) does it *discriminate* skill presence at the line level? Each question is necessary but not sufficient alone. A method could rank episodes perfectly (high r) but with a wildly wrong scale (negative Exp.Var(raw)); conversely, a method could fit the scale correctly in-sample but collapse under CV. The six metrics are therefore complementary, not redundant.

4.3.1 Train Pearson r: Rank-Order Agreement

What it measures. Pearson r measures the linear association between predicted EPSPI_e and observed $\bar{Y}_e$ across all 49 episodes. A value of $r = 1$ means the method perfectly ranks episodes by their skill deployment; $r = 0$ means EP structure carries no signal.

Why use it. Pearson r is the standard measure of predictive validity in educational measurement [24] and has been used as the primary metric in studies relating pedagogical features to learning outcomes [15, 18]. It is interpretable: r^2 is the proportion of variance explained (equivalent to R^2 in simple linear regression). In our context it answers the pedagogical question directly: do lessons coded with more skill-conducive EPs actually produce more skill-present moments?

Worked example. Suppose five episodes have the following values (a subset of $E = 49$):

Episode	EPSPI_e (predicted)	$\bar{Y}_e$ (observed)
Ep 3	0.12	0.15
Ep 7	0.16	0.19
Ep 13	0.51	0.61
Ep 21	0.14	0.11
Ep 38	0.18	0.22

The rankings are almost identical (Ep13 highest in both, Ep3/21 lowest), giving a high r. Now suppose a broken method predicts $\text{EPSPI}_e = 0.15$ for every episode regardless of EP structure. Then $r = 0$ because a constant prediction has zero variance and pearsonr is

undefined (or zero by convention). This is exactly the failure mode of methods with low $P(\mathrm{EP}_{\mathrm{Con}})$ signal.

Limitation. Train r is computed on all 49 episodes including those used to estimate θ_{ij}. It therefore measures in-sample fit, not predictive validity on new episodes. CV r (below) corrects for this.

4.3.2 Exp.Var(cal) vs Exp.Var(raw): Scale vs Rank

What they measure. Both measure how much of the variance in $\bar{Y}_e$ is explained, but they differ in whether scale is penalised.

Exp.Var(raw) uses the raw EPSPI predictions without any rescaling:

$$\mathrm{Exp.Var(raw)} = 1 - \frac{\mathrm{Var}(\bar{Y}_e - \mathrm{EPSPI}_e)}{\mathrm{Var}(\bar{Y}_e)}.$$

If EPSPI_e is systematically too low or too high (scale bias), the residuals $\bar{Y}_e - \mathrm{EPSPI}_e$ are large even if the ranking is correct, and Exp.Var(raw) can be negative.

Exp.Var(cal) first fits an OLS calibration $\hat{Y}_e = a + b \cdot \mathrm{EPSPI}_e$ and computes:

$$\mathrm{Exp.Var(cal)} = 1 - \frac{\mathrm{Var}(\bar{Y}_e - \hat{Y}_e^{\mathrm{cal}})}{\mathrm{Var}(\bar{Y}_e)}.$$

This is numerically identical to R^2 from the OLS regression of $\bar{Y}_e$ on EPSPI_e, which equals r^2 for simple regression. Calibration removes scale bias before computing variance explained.

Why use both. The *gap* Exp.Var(cal) − Exp.Var(raw) is a direct diagnostic of scale bias [11]. Methods C (PCA) and G (NMF) produce $\mathrm{EPSPI}_e \approx 0.03$–$0.05$ because the principal component or NMF factor is anchored to the Constructivism direction, compressing most episodes near zero. The raw predictions are so far below $\bar{Y}_e \approx 0.20$ that Exp.Var(raw) is *negative* (the method is worse than predicting the mean), yet Exp.Var(cal) is positive (≈ 0.49) because the *ranking* is correct. Reporting only Exp.Var(cal) would hide this scale failure; reporting only Exp.Var(raw) would penalise a method that is conceptually sound but needs rescaling for deployment. Both together give the complete picture.

Worked example. Using the same five episodes with Method C (PCA):

Ep	$\bar{Y}_e$	EPSPI_e^C (raw)	$\hat{Y}_e^{\text{cal}}$	Raw residual
Ep 3	0.150	0.021	0.148	−0.129
Ep 7	0.190	0.038	0.191	−0.152
Ep 13	0.610	0.180	0.608	−0.430
Ep 21	0.110	0.015	0.112	−0.095
Ep 38	0.220	0.044	0.218	−0.176

The raw residuals are large (method predicts ∼0.02–0.18 when true values are 0.11–0.61), giving negative Exp.Var(raw). But the *calibrated* predictions $\hat{Y}_e^{\text{cal}}$ track $\bar{Y}_e$ closely, giving high Exp.Var(cal). The OLS slope $b \approx 3.2$ rescales PCA predictions into the correct range — the information is there, just on the wrong scale.

4.3.3 CV Pearson r: Out-of-Sample Predictive Validity

What it measures. CV r is the mean Pearson correlation between predicted and observed $\bar{Y}_e$ on *held-out* episodes across five folds. It estimates how well an EPSPI method would predict skill deployment in a new episode never seen during fitting.

Why use it. In-sample metrics (Train r, Exp.Var) can be inflated by overfitting, especially for methods with many parameters (RF, SVM, GEE with separate coefficients per skill). CV r is the gold standard for predictive validity in educational data mining [3, 26]. The episode-grouped split is essential: if lines from the same episode appeared in both train and test folds, the θ_{ij}^e matrix would leak ground-truth skill rates into the features, inflating CV r artificially [20, 19]. GroupKFold at the episode level guarantees no such leakage.

Worked example: what a fold looks like. With $E = 49$ and 5 folds, each test fold contains approximately 10 episodes (≈ 3,836 lines on average).

- **Fold 2 test:** Episodes {4, 9, 14, 20, 27, 33, 38, 44, 48, 49}
- **Fold 2 train:** all remaining 39 episodes

The method (e.g. GEE) is re-estimated from scratch on the 39 training episodes. New θ_{ij}^e matrices are computed for the 10 test episodes using only those episodes' own lines (no training data leaks in). EPSPI_e is computed for each test episode, compared to its $\bar{Y}_e$, and Pearson r is recorded. This repeats for all 5 folds; the mean is CV r.

Relationship to Train r. The drop from Train $r \approx 0.70$ to CV $r \approx 0.45$ is expected and healthy. A drop of 0.25 is not evidence of overfitting — it reflects the genuine loss of in-sample θ_{ij}^e calibration when predicting new episodes. If CV $r \approx$ Train r, the method

has very low variance (e.g. Method A, which uses no learned parameters beyond θ). If CV $r \ll$ Train r (e.g. SVM: $0.85 \to 0.31$), the method overfits the 49-episode training set.

4.3.4 CV MSE: Magnitude of Prediction Error

What it measures. CV MSE measures the average squared prediction error per episode on held-out folds:

$$\text{CV MSE} = \frac{1}{5}\sum_{k=1}^{5}\frac{1}{|F_k|}\sum_{e\in F_k}\big(\text{EPSPI}_e - \bar{Y}_e\big)^2.$$

Unlike CV r, which is scale-invariant, MSE penalises both rank errors and scale errors simultaneously.

Why use it alongside CV r. CV r can be high while MSE is also high if predictions are monotonically correct but far from the true values in magnitude. MSE is the standard loss function for regression tasks [14] and directly measures operational usefulness: a teacher or curriculum designer who uses EPSPI_e as an absolute estimate of expected skill rate needs low MSE, not just high rank-order correlation.

Worked example. Method D (Logistic) achieves CV $r = 0.400$ and CV MSE $= 0.008525$. Method C (PCA) achieves CV $r = 0.281$ but CV MSE $= 0.042322$ — five times larger. PCA's scale compression (predictions near 0.04 when true values are $\sim$0.20) inflates MSE even though the ranking is partially preserved. The GEE method achieves the best CV $r = 0.450$ *and* the lowest CV MSE $= 0.008018$, confirming it as the overall best method.

4.3.5 AIC and BIC: Line-Level Discriminative Power

What they measure. For each method, a fixed external logistic regression is fitted at the line level:

$$\text{logit}\big(P(Y_{nj}=1)\big) = \beta_{0j} + \beta_{1j}\cdot\text{EPSPI}_n, \quad j = 1,\ldots,4.$$

AIC and BIC are computed from the summed log-likelihood of this model across all four skills:

$$\text{AIC} = \sum_{j=1}^{4}(2k_j - 2\ell_j), \qquad \text{BIC} = \sum_{j=1}^{4}(k_j \ln N - 2\ell_j),$$

where $k_j = 2$ (intercept + slope) and $N = 19{,}181$. Lower AIC/BIC means the logistic model fits better with that method's EPSPI_n as the predictor.

Why use AIC/BIC for non-parametric methods. AIC [2] and BIC [27] are normally used to compare parametric models of different complexity. Here they are applied consistently across all 11 methods — including non-parametric ones (RF, SVM, NMF)

— because the *likelihood comes from the fixed external logistic model*, not from the EPSPI method itself. Since $k_j = 2$ and $N = 19{,}181$ are identical for all methods, AIC differences are log-likelihood differences exclusively. This provides a principled line-level comparison without assuming the methods are parametric [7].

Worked example: what AIC differences mean. The Correspondence Analysis (CA) method achieves the lowest AIC (44,358), while the Random Forest achieves the highest (48,508). The difference $\Delta\text{AIC} = 4{,}150$ is decisive on the **(author?)** [7] scale (differences > 10 are considered very strong evidence against the higher-AIC model). CA wins AIC because it directly optimises the chi-squared EP–skill co-occurrence structure at the line level — exactly what the logistic likelihood rewards. However, CA achieves only CV $r = 0.296$ at the episode level, revealing a tension: line-level discriminative power (AIC) and episode-level predictive validity (CV r) measure different aspects of performance. A method can be excellent at one and poor at the other. Reporting both together exposes this tension transparently.

Numerical guide. On the **(author?)** [7] scale: $\Delta\text{AIC} < 2$ = essentially equivalent; 2–7 = moderate evidence for lower-AIC method; > 10 = decisive. In our results, all pairwise $\Delta\text{AIC} > 100$, meaning the differences are unambiguous.

4.3.6 Why Six Metrics and Not One

No single metric captures the full picture [30, 9]:

Metric	Question answered	Blind spot
Train r	Rank agreement in-sample	Scale; overfitting
Exp.Var(cal)	Rank + variance in-sample	Scale bias hidden
Exp.Var(raw)	Absolute scale agreement	Penalises good rankers
CV r	Out-of-sample ranking	Scale; magnitude
CV MSE	Out-of-sample magnitude	Rank order
AIC/BIC	Line-level discrimination	Episode-level aggregation

The GEE method is the recommended choice because it achieves the best CV r (0.450) and the lowest CV MSE (0.008018) simultaneously. AIC and Exp.Var(raw) are reported for completeness and to make the scale-bias pattern transparent.

5 Eleven Estimation Methods

6 Estimation Methods

We organise the eleven frequentist formulations of EPSPI along three axes of statistical inquiry, each reflecting a distinct scientific question about the EP–skill relationship (Table 4). The first axis — **direct estimation** — asks what value $\hat{\theta}_{ij}$ takes under progressively stronger modelling assumptions [8]. Methods A and B establish a non-parametric baseline using raw conditional frequencies [1], differing only in whether EP weights are equal or frequency-proportional. Method D introduces the first parametric assumption — that the log-odds of skill presence is linear in EP indicators — via logistic regression [16], and Method F extends this with a within-episode clustering correction through GLMM random intercepts [23, 25]. Method GEE completes this axis by relaxing the within-episode independence assumption without specifying a full likelihood, estimating population-average EP–skill associations robust to within-episode correlation [22, 32]. The second axis — **representation learning** — asks whether the $P \times K = 12$-dimensional EP–skill profile space contains latent structure that improves estimation of $\hat{\theta}_{ij}$ [4]. Method C (PCA) finds linear components of maximum variance without constraints [17], Method CA (Correspondence Analysis) discovers the principal axes of EP–skill co-occurrence in a contingency-table framework [12], and Method G (NMF) adds the domain-appropriate non-negativity constraint, discovering additive teaching-mode profiles that reconstruct each episode's EP–skill fingerprint [21]. The third axis — **predictive modelling** — asks how accurately skill expression can be predicted from EP patterns using methods that differ in their assumptions about the data-generating process [6]. Method E (Naïve Bayes) inverts the conditional probability via Bayes' theorem under a conditional-independence assumption [13], Method RF (Random Forest) learns non-linear EP–skill interactions through an ensemble of decorrelated trees [5], and Method K (SVM) abandons distributional assumptions entirely, seeking instead the maximum-margin boundary in a kernel-induced feature space [29, 10].

Together, the eleven methods span the major schools of frequentist statistical thought — non-parametric, parametric, hierarchical, robust, latent-variable, generative, and discriminative — applied to a common estimand: the EP–skill affinity matrix $\hat{\theta}_{ij} = \hat{P}(S_j \mid \mathrm{EP}_i)$. This breadth serves two purposes: it demonstrates that substantive conclusions about EP–skill relationships are robust across methodological choices [28], and it reveals where and why methods diverge, illuminating the structure of the EPSPI estimation problem itself.

Methods are organised along three axes:

Table 4: Organisation of the eleven EPSPI estimation methods along three axes of statistical inquiry.

Axis	**Methods**	**Characteristic**
Direct Estimation	A, B, D, F, GEE	Model $P(S_j \mid \mathrm{EP}_i)$ explicitly under progressively stronger parametric assumptions
Representation Learning	C, CA, G	Reduce the $P \times K$ EP–skill space to latent dimensions; $\hat{\theta}_{ij}$ is recovered from low-rank structure
Predictive Modelling	E, RF, K	Treat EPSPI_n as a regression target; discriminative models trained to minimise prediction error

Axis	Methods	Characteristic
Direct Estimation	A, B, D, F, GEE	Model $P(S_j \mid \mathrm{EP}_i)$ explicitly
Representation Learning	C, CA, G	Reduce EP–skill space to latent dimensions
Predictive Modelling	E, RF, K	Discriminative, treat EPSPI as target

6.1 Method A: Simple Average

Method A is the direct plug-in estimator of EPSPI under equal weighting of all active EPs and all skills. It is a **non-parametric estimator**: the only quantities entering the formula are the Laplace-smoothed global conditional probabilities θ_{ij} (Equation (**??**)), computed once from all N lines and fixed thereafter. No parameters are fitted to episode-level outcomes.

Line-level score. Let $p_n = \sum_{i=1}^{P} X_{ni}$ be the number of EPs active in line n. The Method A score for line n is:

$$\mathrm{EPSPI}_n^{(A)} = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni} \left(\frac{1}{K} \sum_{j=1}^{K} \theta_{ij} \right), \qquad p_n > 0, \tag{15}$$

where θ_{ij} is the **global** Laplace-smoothed conditional probability matrix (Equation (**??**)). The indicator X_{ni} selects only the active EPs; dividing by p_n ensures a line with one active EP and a line with three active EPs contribute equally to the episode mean. When $p_n = 0$ — which does not occur in this corpus by annotation design — the fallback $\mathrm{EPSPI}_n^{(A)} = (PK)^{-1} \sum_{i=1}^{P} \sum_{j=1}^{K} \theta_{ij}$ is used, which is the grand mean of Θ.

Deployment example. For a line with only Constructivism active ($X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$, so $p_n = 1$):

$$\mathrm{EPSPI}_n^{(A)} = \frac{1}{4}(0.8061 + 0.0418 + 0.8783 + 0.3118) = 0.5095.$$

A practitioner needs only the published Θ table (Table **??**) and their own EP binary flags X_{ni} — no skill observations Y_{nj} are required.

Episode-level aggregation. The episode-level score is the unweighted mean over the n_e lines belonging to episode e:

$$\mathrm{EPSPI}_e^{(A)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(A)}, \tag{16}$$

where $n_e = |\{n : e_n = e\}|$ is the actual number of lines in episode e. No equal-size assumption is imposed across episodes.

Why global θ_{ij} and not episode-conditional $\theta_{ij}^{e_n}$. Method A deliberately uses the global θ_{ij} for two reasons. First, $\theta_{ij}^{e_n}$ is estimated from the skill observations Y_{nj} of the *same* episode used to compute the validation target $\bar{Y}_e$, introducing a direct within-episode dependency between predictor and outcome that the cross-validated evaluation is designed to prevent. Second, Method A is a **non-parametric plug-in estimator**: it substitutes observed conditional frequencies directly into the formula without fitting any functional form to Y_{nj}. Methods E, G, and I also use episode-conditional information but do so through a parametric likelihood fitted jointly across all episodes — their parameters $\hat{\beta}$ are estimated from training episodes and applied to test episodes where Y_{nj} is never observed during fitting, avoiding the same circularity. The CV gap between Methods A/B (global θ) and E/G/I (episode-conditional estimation) therefore quantifies the predictive gain from incorporating within-episode skill information under proper cross-validation.

Properties. Method A has **zero free parameters**. Because p_n weights each active EP equally regardless of its corpus frequency, Method A does not distinguish an episode dominated by a single rare EP (e.g. pure Constructivism, $\delta_{\mathrm{Con}}^e = 1.00$) from one where that EP co-occurs with others at equal frequency — a limitation addressed by Method B, which replaces the line-level X_{ni} weights with episode-level proportions δ_i^e.

6.2 Method B: δ-Weighted Average

Method A weights each active EP by its line-level activation X_{ni}, normalised by p_n. Method B replaces this with the **episode EP proportion** δ_i^e — how dominant each pedagogical paradigm is across the whole episode. A line embedded in a predominantly Constructivist

episode receives a higher Constructivism weight than the same line in a mixed episode, even if its own EP flags are identical.

Step 1 — Episode EP proportions δ_i^e. For each episode e and EP i:

$$\delta_i^e = \frac{1}{n_e} \sum_{\{n\,:\,e_n=e\}} X_{ni}, \tag{17}$$

where n_e is the number of lines in episode e. $\delta_i^e \in [0,1]$ is the proportion of lines in which EP_i is active. Because multiple EPs may be simultaneously active in the same line, $\sum_{i=1}^{P} \delta_i^e$ can exceed 1. δ_i^e requires only X_{ni} — no skill observations Y_{nj} — so it can be computed before or without skill coding, carrying no leakage at deployment.

Example: Episode 7. Using the six-line episode from Section **??** (Table **??**), recall the EP annotation counts:

	Beh ($i=1$)	Cog ($i=2$)	Con ($i=3$)
$\sum_{\{n:e_n=7\}} X_{ni}$	3	3	3
$n_e = 6$			
$\delta_i^{e=7}$	$3/6 = 0.500$	$3/6 = 0.500$	$3/6 = 0.500$

In this balanced toy episode all three EPs appear equally often, so $\delta_i^{e=7} = 0.5$ for all i. In a real episode such as Episode 13 (pure Constructivism), $\delta_{\mathrm{Con}}^{13} = 1.00$ and $\delta_{\mathrm{Beh}}^{13} = \delta_{\mathrm{Cog}}^{13} = 0.00$, which is why Episode 13 is a high-leverage point — its δ profile is an extreme vertex of the simplex.

Step 2 — Line-level score. The Method B score for line n uses $\delta_i^{e_n}$ as the EP weight and the row mean of Θ as the skill summary:

$$\mathrm{EPSPI}_n^{(B)} = \sum_{i=1}^{P} \delta_i^{e_n} \left(\frac{1}{K} \sum_{j=1}^{K} \theta_{ij} \right), \tag{18}$$

where θ_{ij} is the global Laplace-smoothed conditional probability (Equation (**??**)). The inner average $\frac{1}{K}\sum_j \theta_{ij}$ is the i-th row mean of Θ — no new symbol is required.

Example: Episode 7. Using $\delta_i^{e=7} = 0.5$ for all i and the global Θ matrix (Table **??**):

$$\frac{1}{K}\sum_j \theta_{\text{Beh},j} = \tfrac{1}{4}(0.3239 + 0.0071 + 0.0360 + 0.0982) = 0.116$$
$$\frac{1}{K}\sum_j \theta_{\text{Cog},j} = \tfrac{1}{4}(0.2138 + 0.0069 + 0.0094 + 0.3591) = 0.147$$
$$\frac{1}{K}\sum_j \theta_{\text{Con},j} = \tfrac{1}{4}(0.8061 + 0.0418 + 0.8783 + 0.3118) = 0.510$$

Every line in Episode 7 receives the same score:

$$\text{EPSPI}_n^{(B)} = 0.5 \times 0.116 + 0.5 \times 0.147 + 0.5 \times 0.510 = 0.058 + 0.074 + 0.255 = \mathbf{0.387}$$

Compare this to the Method A scores for the same episode, which varied from 0.286 (Beh-only line) to 0.357 (Con-only line): Method B replaces that within-episode variation with a single episode-constant value, trading granularity for deployability.

Episode-level aggregation. Because $\text{EPSPI}_n^{(B)}$ is identical for all n_e lines within episode e, the episode mean collapses to:

$$\text{EPSPI}_e^{(B)} = \frac{1}{n_e}\sum_{\{n\,:\,e_n=e\}} \text{EPSPI}_n^{(B)} = \sum_{i=1}^{P} \delta_i^e \left(\frac{1}{K}\sum_{j=1}^{K}\theta_{ij}\right), \tag{19}$$

which is the dot product of the episode EP profile $[\delta_1^e, \delta_2^e, \delta_3^e]$ with the fixed row-mean vector of Θ.

What the evaluation metrics mean for Methods A and B. Methods A and B have **zero free parameters**: θ_{ij} is computed directly from frequency counts and fixed before any evaluation. Because no parameters are fitted to the 49 episodes, the distinction between "training" and "testing" does not apply in the conventional sense. The six evaluation metrics therefore have the following interpretations for these methods:

- **In-sample Pearson** r (labelled Train r in the results table) is the Pearson correlation between $\text{EPSPI}_e^{(A/B)}$ and $\bar{Y}_e$ computed over all 49 episodes. There is no overfitting risk because no episode-level parameters were estimated; the label "train" is a misnomer and should be read as "in-sample association."
- **Exp.Var(cal)** is the R^2 from an OLS regression of $\bar{Y}_e$ on $\text{EPSPI}_e^{(A/B)}$ over all 49 episodes. The OLS calibration ($\hat{Y}_e = a + b \cdot \text{EPSPI}_e$) uses 2 parameters but these are not part of the EPSPI estimator itself — they are a post-hoc rescaling used only for evaluation.

- **Exp.Var(raw)** measures the same thing without the OLS rescaling, penalising scale mismatch directly.
- **CV Pearson** r is the only metric where the zero-parameter nature of A and B genuinely matters. In each fold, θ_{ij} is recomputed from the 39 training episodes only, then frozen; $\text{EPSPI}_e^{(A/B)}$ is computed for the 10 test episodes using this held-out θ_{ij} estimate. This measures whether the EP–skill affinity estimated from 39 episodes generalises to 10 unseen ones. For zero-parameter methods, the expected drop from in-sample r to CV r is small — only the variance in θ_{ij} estimation across folds contributes, not model overfitting.
- **CV MSE** follows the same fold structure as CV r and measures the magnitude of prediction error on held-out episodes.
- **AIC/BIC** are computed from a fixed external logistic model at the line level ($\text{logit}\, P(Y_{nj} = 1) = \beta_{0j} + \beta_{1j} \cdot \text{EPSPI}_n^{(A/B)}$, $k_j = 2$). This model has its own 2 parameters per skill (fitted at the line level), but these are the evaluator's parameters, not Method A/B's. AIC/BIC therefore measure how well $\text{EPSPI}_n^{(A/B)}$ discriminates skill presence at the line level.

For methods with free parameters (E, G, I, J, K), the in-sample metrics reflect genuine overfitting risk and the Train/CV gap is substantively meaningful. For Methods A and B, the Train/CV gap reflects only estimation variance in θ_{ij} across the 5 CV folds.

Properties. Method B has zero free parameters and is fully deployable from episode-level EP records alone. Its key limitation is that $\text{EPSPI}_n^{(B)}$ is episode-constant: all lines within episode e receive the same score, so within-episode EP dynamics — a lesson that shifts from Behaviourism to Constructivism mid-episode — are invisible to Method B but captured by Method A's line-varying X_{ni}/p_n weights.

6.3 Method D: Logistic Regression

Four separate logistic regression models are fitted, one per skill:

$$P(Y_{nj} = 1 \mid X_{n\cdot}) = \sigma\left(\beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni}\right),$$

where $\sigma(z) = (1 + e^{-z})^{-1}$. The index averages predicted probabilities across skills:

$$\text{EPSPI}_n^{(D)} = \frac{1}{K} \sum_{j=1}^{K} \sigma\left(\hat{\beta}_{0j} + \sum_{i=1}^{P} \hat{\beta}_{ij} X_{ni}\right). \tag{20}$$

Estimated coefficients from the full dataset are given in Table 5.

Table 5: Logistic regression coefficients $\hat{\beta}_{ij}$ (Method D). Intercepts $\hat{\beta}_{0j}$ are omitted for brevity.

Skill	$\hat{\beta}_{\text{Beh}}$	$\hat{\beta}_{\text{Cog}}$	$\hat{\beta}_{\text{Con}}$
Problem Solving	0.9652	0.6467	—
Creativity	0.6467	—	—
Collaborative Learning	1.3043	—	large†
Critical Thinking	-1.3641	—	—

†Near-perfect separation; see GEE quasi-separation note.

6.4 Method E: Logistic Regression

Methods A and B estimate $P(S_j \mid \text{EP}_i)$ as a raw conditional frequency — the proportion of lines with both EP i and skill j active. Method E replaces this non-parametric plug-in with a **parametric logistic model**, assuming that the log-odds of skill S_j being present is a linear function of the EP binary indicators. This introduces the first functional-form assumption in the EPSPI framework, at the cost of $K(P+1) = 16$ free parameters.

Model specification. For each skill $j \in \{1, \ldots, K\}$, a binary logistic regression is fitted at the line level:

$$\operatorname{logit} \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) = \beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni}, \tag{21}$$

where β_{0j} is the skill-specific intercept and β_{ij} is the log-odds change in skill S_j associated with EP_i being active. The model is fitted separately for each of the $K = 4$ skills, yielding a $P \times K$ coefficient matrix $B = [\beta_{ij}]$ and a K-vector of intercepts $[\beta_{0j}]$. Estimation uses maximum likelihood over all $N = 19{,}181$ lines.

The predicted probability of skill S_j for line n is:

$$\hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) = \sigma\left(\beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni}\right), \tag{22}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic function.

Relationship to θ_{ij}. In Method A, the conditional probability of skill S_j given only EP_i active ($p_n = 1$, $X_{ni} = 1$, $X_{ni'} = 0$ for $i' \neq i$) is directly θ_{ij}. In Method E, the same quantity is:

$$\hat{P}(Y_{nj} = 1 \mid X_{ni} = 1,\, X_{ni'} = 0\ \forall i' \neq i) = \sigma(\beta_{0j} + \beta_{ij}). \tag{23}$$

Method E therefore implicitly defines a parametric analogue of θ_{ij}:

$$\tilde{\theta}_{ij}^{(E)} = \sigma(\beta_{0j} + \beta_{ij}),$$

constrained to follow the logistic functional form, unlike the unconstrained frequency estimate θ_{ij} of Method A. When EP co-occurrence is present ($p_n > 1$), Method E captures the joint effect through the additive log-odds structure, whereas Method A averages θ_{ij} values independently.

Line-level score. The Method E score for line n is the mean predicted skill probability across all K skills:

$$\mathrm{EPSPI}_n^{(E)} = \frac{1}{K} \sum_{j=1}^{K} \sigma\left(\beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni} \right). \tag{24}$$

The full EP flag vector $X_{n\cdot}$ enters the formula — not just the active EPs — because the logistic model predicts skill probability for any combination of EP flags simultaneously. The p_n normalisation of Method A is therefore not needed: multi-EP lines are handled naturally through the additive log-odds.

Deployment example. For a line with only Constructivism active ($X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$):

$$\mathrm{EPSPI}_n^{(E)} = \frac{1}{4} \sum_{j=1}^{4} \sigma(\beta_{0j} + \beta_{3j}).$$

A practitioner needs only the published coefficient vectors $[\beta_{0j}, \beta_{1j}, \beta_{2j}, \beta_{3j}]$ for each skill j and their own EP binary flags X_{ni} — the same input requirement as Method A, but the published parameters are regression coefficients rather than a frequency table.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(E)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(E)}, \tag{25}$$

where n_e is the actual number of lines in episode e.

What the evaluation metrics mean for Method E. Unlike Methods A and B, Method E has $K(P+1) = 4 \times 4 = 16$ free parameters fitted to Y_{nj} via maximum likelihood. The evaluation metrics therefore carry a different interpretation:

- **In-sample Pearson r and Exp.Var(cal)** reflect genuine in-sample fit of a parametric model. Overfitting risk is small — 16 parameters on 19,181 lines is a highly over-determined system — but nonzero. The label "Train r" is accurate here, unlike Methods A/B where it was a misnomer for in-sample association.
- **CV Pearson r and CV MSE** are the primary reliability metrics. In each fold the 16 coefficients are re-estimated from ∼15,000 training lines and applied to ∼4,000

test lines whose Y_{nj} was never seen during fitting. The Train/CV gap for Method E reflects both estimation variance and genuine model overfitting — in contrast to Methods A/B where only θ_{ij} estimation variance contributes.

- **AIC/BIC** use $k = K(P+1) = 16$ in the penalty, versus $k = 0$ for Methods A and B. Lower AIC despite the heavier penalty indicates genuine log-likelihood improvement from the logistic functional form over raw frequency estimation.

Quasi-separation and Constructivism. Constructivism is active in only 1.1% of lines ($\sum_n X_{n,\text{Con}} \approx 211$) yet has the highest skill co-occurrence rates in Θ ($\theta_{\text{Con,PS}} = 0.806$, $\theta_{\text{Con,CL}} = 0.878$). This extreme imbalance causes **quasi-complete separation** in the logistic model for Problem Solving and Collaborative Learning: the 211 Constructivist lines nearly perfectly predict high skill presence, driving $\hat{\beta}_{3j} \to \infty$. This is handled by L2 (ridge) regularisation with $\lambda = 1.0$, which shrinks the Constructivism coefficients toward zero and produces finite, stable estimates at the cost of slight underestimation of $\hat{P}(S_j = 1 \mid \text{Con})$ relative to the raw frequency θ_{ij}. Method G (GLMM) provides a robustness check by fitting the same logistic structure with random episode intercepts, testing whether the fixed-effect estimates of Method E are distorted by within-episode correlation.

Properties. Method E has $K(P+1) = 16$ free parameters and is the first parametric method in the framework. Its primary limitation is the additive log-odds assumption: the effect of EP_i on $\operatorname{logit} P(S_j = 1)$ is assumed independent of which other EPs are simultaneously active, which may not hold when EP co-occurrence is pedagogically meaningful. Method G relaxes the within-episode independence assumption by adding random intercepts; Method I (GEE) retains the linear log-odds structure but replaces the likelihood with a robust estimating equation, removing the need to specify the within-episode correlation structure entirely.

6.5 Method F: GLMM

A Binomial Bayesian Mixed GLMM with a random intercept per episode is fitted for each skill:

$$\operatorname{logit}\big(P(Y_{nj} = 1 \mid X_{n\cdot}, u_e)\big) = \beta_{0j} + \sum_i \beta_{ij} X_{ni} + u_{e_n}, \quad u_e \sim \mathcal{N}(0, \tau^2).$$

For prediction, the marginal (population-average) value $u_e = 0$ is used, making the GLMM prediction function identical to Method D. The primary output is the **intraclass correlation coefficient**:

$$\text{ICC} = \frac{\tau^2}{\tau^2 + \pi^2/3},$$

which quantifies the proportion of skill variance attributable to episode-level clustering. Method F is a robustness check for D; CV is not performed ($\approx$ 20 minutes to refit 25 times with no new information beyond Method D).

6.6 Method F: Generalised Linear Mixed Model (GLMM)

Method E fits a fixed-effects logistic regression at the line level, treating all $N = 19{,}181$ lines as independent. This assumption is violated: lines within the same episode share a common classroom context — teacher style, lesson topic, student cohort — that induces within-episode correlation in skill observations Y_{nj}. Ignoring this correlation does not bias the fixed-effect estimates $\hat{\beta}_{ij}$ but does underestimate their standard errors, inflating apparent precision [25, 23]. Method F addresses this by adding an **episode-level random intercept** to the logistic model, explicitly modelling the between-episode variance in baseline skill rates.

Model specification. For each skill S_j, the GLMM is:

$$\operatorname{logit} P(Y_{nj} = 1 \mid X_{n\cdot},\, u_{ej}) = \underbrace{\beta_{0j} + \sum_{i=1}^{P} \beta_{ij}\, X_{ni}}_{\text{fixed effects}} + \underbrace{u_{ej}}_{\text{random intercept}}, \tag{26}$$

where $u_{ej} \overset{\text{iid}}{\sim} \mathcal{N}(0,\, \sigma_j^2)$ is the random intercept for episode e and skill j, capturing unobserved episode-level heterogeneity in skill baseline rates. The random intercept u_{ej} is not a free parameter to be estimated per episode — it is a latent variable integrated out during maximum likelihood estimation via Gauss–Hermite quadrature [**?**]. The free parameters are $\{\beta_{0j}, \beta_{1j}, \beta_{2j}, \beta_{3j}, \sigma_j^2\}$ for each skill j, giving $(P+2) \times K = 20$ free parameters in total ($P + 1$ fixed effects plus 1 variance component per skill).

Relationship to Method E. Method F is a direct extension of Method E. The fixed-effect parameters β_{ij} in Method F have the same interpretation as in Method E — log-odds contribution of EP_i to skill S_j — but they are now estimated *conditional on* the random intercept u_{ej}, making them **subject-specific** (conditional) rather than **population-average** (marginal) effects [32]. In practical terms:

- Method E $\hat{\beta}_{ij}$: the average log-odds increase in $P(Y_{nj} = 1)$ when EP_i is activated, averaged over all episodes.
- Method F $\hat{\beta}_{ij}$: the log-odds increase for a *specific episode* holding its random intercept u_{ej} fixed — a larger effect size than the marginal estimate because it is not attenuated by between-episode variance.

The intraclass correlation coefficient (ICC) quantifies the proportion of total skill variance attributable to between-episode differences:

$$\mathrm{ICC}_j = \frac{\sigma_j^2}{\sigma_j^2 + \pi^2/3}, \tag{27}$$

where $\pi^2/3 \approx 3.29$ is the variance of the standard logistic distribution [**?**]. A high ICC (e.g. ICC > 0.10) indicates that episode-level clustering is substantial and that ignoring it in Method E materially affects inference.

Line-level score. At deployment, the random intercept u_{ej} is an unobserved latent variable. The marginal predicted probability for line n, integrating out u_{ej}, does not have a closed form for the logistic link. Following standard practice [23], we use the **marginal prediction** obtained by setting $u_{ej} = 0$ (the population mean):

$$\mathrm{EPSPI}_n^{(F)} = \frac{1}{K} \sum_{j=1}^{K} \sigma\left(\hat{\beta}_{0j} + \sum_{i=1}^{P} \hat{\beta}_{ij}\, X_{ni} \right), \tag{28}$$

which is structurally identical to Equation (24) for Method E, with the difference that the $\hat{\beta}_{ij}$ values are now estimated under the GLMM likelihood that accounts for within-episode clustering. Setting $u_{ej} = 0$ gives the prediction for an "average episode" — a new episode drawn from the population with no prior knowledge of its specific classroom context.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(F)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(F)}, \tag{29}$$

where n_e is the actual number of lines in episode e. Because u_{ej} is set to zero in Equation (56), lines within the same episode that share identical EP flags produce identical $\mathrm{EPSPI}_n^{(F)}$ values. Within-episode variation in $\mathrm{EPSPI}_n^{(F)}$ arises only from line-to-line variation in the EP profile $[X_{n1}, X_{n2}, X_{n3}]$.

Why Method F is in-sample only. Method F uses `BinomialBayesMixedGLM` from `statsmodels` [**?**], which fits the full random-effects likelihood via Laplace approximation with Gaussian priors on the fixed effects. Refitting this model 25 times (5 folds $\times$ 5 CV repeats) would require approximately 20 minutes of computation with no expected gain in information: because the marginal prediction sets $u_{ej} = 0$, the CV prediction function for Method F is structurally identical to that of Method E ($u_{ej} = 0$ in both). The CV Pearson r for Method F is therefore expected to equal that of Method E (≈ 0.400) to within Monte Carlo noise, and Method F is reported as *in-sample only* in the results

table. Its contribution to the paper is not predictive performance but the ICC values, which quantify the episode-level clustering that motivates the GEE working-correlation specification in Method I.

Quasi-separation and convergence guard. As in Method E, Constructivism's sparsity (1.1% of lines) causes quasi-complete separation in the logistic likelihood [**?**]. A convergence guard $|\hat{\beta}_{ij}| \leq 50$ is applied; coefficients hitting this bound are flagged in the results. Method I (GEE) addresses quasi-separation more principally through the working correlation structure and robust sandwich standard errors, without requiring a hard coefficient bound.

What "Train Pearson r" means for Method F. Method F has $(P + 2) \times K = 20$ free parameters estimated on all N lines. The in-sample Train Pearson r reflects genuine in-sample fit under the GLMM likelihood, with the same caveat as Method E: the risk of overfitting is negligible at the line level ($N = 19{,}181 \gg 20$ parameters) but meaningful at the episode level (49 episodes). Because the marginal prediction $u_{ej} = 0$ is used, the episode-level EPSPI for Method F is expected to be close to that of Method E, and any difference in Train r or Exp.Var between the two methods reflects the change in $\hat{\beta}_{ij}$ induced by accounting for within-episode clustering — not a change in the prediction function's structure.

Properties. Method F has $(P + 2) \times K = 20$ **free parameters** — four more than Method E, corresponding to the $K = 4$ random-effect variance components σ_j^2. It is the only method in the framework that explicitly models within-episode skill clustering. Its primary role is to confirm that Method E's fixed-effect estimates $\hat{\beta}_{ij}$ are not materially distorted by ignoring within-episode correlation; if Method E and Method F produce similar EPSPI_e values and ICC values are low (< 0.05), within-episode clustering is not a confound for the other methods. Method G (NMF) and Method I (GEE) take complementary approaches to the same clustering problem through latent-variable decomposition and working-correlation robustness respectively.

6.7 Method I: Generalised Estimating Equations (GEE)

Method E estimates population-average EP–skill associations by treating all N lines as independent. Method F corrects for within-episode clustering through a subject-specific random intercept, but the marginal prediction still sets $u_{ej} = 0$, making its deployment prediction identical in structure to Method E. Method I takes a third approach: it directly models the **within-episode correlation structure** of the binary skill observations Y_{nj} through a working correlation matrix, producing population-average coefficient estimates that are consistent even if the working correlation is mis-specified [22, 32]. GEE does

not require a full likelihood — it solves a set of generalised estimating equations that generalise the score equations of logistic regression to the correlated data setting [**?**].

Model specification. For each skill S_j, GEE specifies the same marginal mean model as Method E:

$$\operatorname{logit} \mu_{nj} = \operatorname{logit} E[Y_{nj} \mid X_{n\cdot}] = \beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni}, \tag{30}$$

with the additional specification of a **working correlation matrix** $\mathbf{R}(\alpha_j)$ for the n_e binary outcomes within episode e:

$$\operatorname{Corr}(Y_{nj}, Y_{n'j} \mid e_n = e_{n'} = e) = \alpha_j, \quad n \neq n', \tag{31}$$

the exchangeable structure, which assumes all pairs of lines within the same episode have the same correlation α_j. The exchangeable structure is chosen because there is no natural ordering of lines within an episode that would justify an autoregressive or unstructured matrix. GEE estimates $\hat{\beta}_{ij}$ and $\hat{\alpha}_j$ iteratively via the **iteratively reweighted least squares** (IRLS) algorithm, alternating between updating $\hat{\beta}_{ij}$ with current $\hat{\alpha}_j$ fixed and updating $\hat{\alpha}_j$ from the Pearson residuals [22]. The final standard errors use the **sandwich (robust) variance estimator** $V_{\text{sand}} = A^{-1}BA^{-1}$, which is consistent regardless of whether the working correlation $\mathbf{R}(\hat{\alpha}_j)$ is correctly specified [**?**].

Relationship to Methods E and F. All three methods — E, F, and I — estimate the same population-average log-odds:

$$\beta_{ij} = \operatorname{logit} P(Y_{nj} = 1 \mid X_{ni} = 1,\, X_{ni'} = 0 \,\forall\, i' \neq i) - \beta_{0j}.$$

They differ in how within-episode correlation is handled during estimation:

- **Method E**: assumes $\alpha_j = 0$ (independence). Consistent if $\alpha_j = 0$; standard errors too small if $\alpha_j > 0$.
- **Method F**: models α_j through a random intercept variance σ_j^2; gives conditional (subject-specific) not marginal estimates.
- **Method I**: estimates α_j directly as a nuisance parameter; gives marginal estimates with sandwich standard errors valid for any α_j.

The estimated within-episode correlation $\hat{\alpha}_j$ is itself a substantive finding: it quantifies how much two randomly chosen lines from the same episode resemble each other in skill expression, above and beyond what is predicted by their EP flags.

Three-stage convergence fallback. Constructivism appears in only 1.1% of lines, causing quasi-complete separation in the GEE score equations — the same problem as Methods E and F. Additionally, the exchangeable working correlation occasionally fails to converge for rare EP profiles. A three-stage fallback is implemented:

1. **Exchangeable** working correlation with $|\hat{\beta}_{ij}| \leq 50$ guard.
2. If exchangeable diverges: **independence** working correlation (reduces to Method E estimates with sandwich standard errors).
3. If independence also diverges: **global θ_{ij} fallback** — the Laplace-smoothed frequency estimate from Equation (**??**).

The fallback stage used for each skill j is recorded and reported in Appendix **??**. For the exchangeable structure, the estimated correlation coefficients across the four skills are: $\hat{\alpha}_{\text{PS}} = 0.450$, $\hat{\alpha}_{\text{Cr}} = 0.043$, $\hat{\alpha}_{\text{CL}} = 0.223$, $\hat{\alpha}_{\text{CT}} = 0.021$, indicating moderate within-episode clustering for Problem Solving and weak clustering for the remaining skills.

Line-level score. The Method I score for line n is the mean marginal predicted probability across all K skills:

$$\text{EPSPI}_n^{(I)} = \frac{1}{K} \sum_{j=1}^{K} \sigma\left(\hat{\beta}_{0j}^{(I)} + \sum_{i=1}^{P} \hat{\beta}_{ij}^{(I)} X_{ni} \right), \tag{32}$$

where $\hat{\beta}_{ij}^{(I)}$ are the GEE-estimated coefficients and $\sigma(\cdot)$ is the logistic function. The formula is structurally identical to Methods E and F (Equations (24) and (56)) — the difference from E and F lies entirely in the coefficient values $\hat{\beta}_{ij}^{(I)}$, which are estimated under the exchangeable working correlation rather than under independence or a random-effects likelihood.

Example: line with only Constructivism active. Setting $X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$:

$$\text{EPSPI}_n^{(I)} = \frac{1}{4} \sum_{j=1}^{4} \sigma\left(\hat{\beta}_{0j}^{(I)} + \hat{\beta}_{3j}^{(I)} \right).$$

The GEE coefficients $\hat{\beta}_{3j}^{(I)}$ will differ from the logistic $\hat{\beta}_{3j}^{(E)}$ because GEE downweights large episodes (high n_e) relative to small episodes when the working correlation is exchangeable — a line in a 587-line episode contributes less to the estimating equations than a line in a 52-line episode, adjusting for the fact that lines in large episodes are more correlated with each other.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(I)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n=e\}} \mathrm{EPSPI}_n^{(I)}, \tag{33}$$

where n_e is the actual number of lines in episode e. As with Methods E and F, $\mathrm{EPSPI}_n^{(I)}$ varies line by line through the EP profile $[X_{n1}, X_{n2}, X_{n3}]$, so the episode mean is a genuine average of heterogeneous line scores rather than a constant.

What "Train Pearson r" means for Method I. Method I has $K \times (P+1) = 16$ free parameters $\hat{\beta}_{ij}^{(I)}$ plus $K = 4$ nuisance correlation parameters $\hat{\alpha}_j$, giving 20 estimated quantities in total. The $\hat{\alpha}_j$ are nuisance parameters — they improve the efficiency of $\hat{\beta}_{ij}^{(I)}$ but are not used in the prediction formula (32). The in-sample Train Pearson r reflects genuine in-sample fit with the same episode-level overfitting caveat as Method E: $\hat{\beta}_{ij}^{(I)}$ is estimated from all 49 episodes' line-level Y_{nj}, so the correlation between $\mathrm{EPSPI}_e^{(I)}$ and $\bar{Y}_e$ is not a held-out evaluation. Method I achieves the highest CV $r = 0.450$ and the lowest CV MSE $= 0.008018$ of all eleven methods, confirming that accounting for within-episode correlation in estimation genuinely improves out-of-sample predictive validity — not merely in-sample fit.

Properties. Method I has $K \times (P+1) = 16$ **free parameters** in the prediction formula, identical in count to Method E, with an additional $K = 4$ nuisance correlation estimates. It achieves the best CV performance of all eleven methods and is therefore the **recommended method** for predicting episode-level skill deployment from EP structure. The key advantages over Method E are: (1) standard errors are valid under any within-episode correlation structure; (2) coefficient estimates are more efficient when $\hat{\alpha}_j > 0$; (3) the sandwich variance estimator provides a robustness check on the logistic functional form assumption. The key advantage over Method F is full cross-validatability: unlike the GLMM likelihood, GEE estimating equations can be solved on 39 training episodes in under one second per fold, making 5-fold CV computationally trivial. Its limitation relative to Method F is that GEE does not provide episode-specific random effect estimates $\hat{u}_{ej}$ and therefore cannot answer questions about individual episode-level deviations from the population average — the non-EP residual $\varepsilon_e = \bar{Y}_e - \mathrm{EPSPI}_e^{(I)}$ serves this diagnostic role instead (Section **??**).

6.8 Method C: Principal Component Analysis

Methods A, B, E, F, and I all work directly with the $P \times K$ conditional probability matrix Θ, using either raw frequencies or parametric log-odds models. Method C takes a different approach: rather than modelling θ_{ij} directly, it asks whether the 12-dimensional EP–skill

Table 6: GEE estimated coefficients $\hat{\beta}^{GEE}$ with robust 95% Wald confidence intervals (logit scale). Asterisks (*) indicate confidence intervals that exclude zero (significant at the 5% level).

Skill	Method	Intercept	$\hat{\beta}_{Beh}$	$\hat{\beta}_{Cog}$	$\hat{\beta}_{Con}$
Problem Solving	Exchangeable	-1.678 [-2.484, -0.873]*	0.655 [0.027, 1.283]*	0.568 [0.033, 1.103]*	1.851 [0.381, 3.322]*
Creativity	Exchangeable	-5.463 [-7.037, -3.890]*	-0.100 [-1.792, 1.592]	0.694 [-0.696, 2.085]	2.076 [0.716, 3.435]*
Collab. Learning	Independence	-5.362 [-7.605, -3.119]*	1.462 [0.629, 2.295]*	-0.516 [-1.073, 0.041]	6.907 [5.136, 8.678]*
Critical Thinking	Exchangeable	-2.129 [-2.928, -1.330]*	-0.984 [-1.256, -0.712]*	1.558 [1.065, 2.051]*	1.436 [0.721, 2.150]*

profile space contains a lower-dimensional **latent structure** that summarises the essential variation in EP–skill co-occurrence patterns. Principal Component Analysis (PCA) finds this structure by identifying the directions of maximum variance in the column space of Θ, without imposing any constraint on the sign or magnitude of the components [17**?**].

Input matrix and centring. PCA is applied to the global Θ matrix. The 3×4 matrix $\Theta = [\theta_{ij}]$ has rows indexed by EP ($i = 1, 2, 3$) and columns indexed by skill ($j = 1, 2, 3, 4$). Before decomposition, Θ is column-centred:

$$\tilde{\theta}_{ij} = \theta_{ij} - \bar{\theta}_{\cdot j}, \qquad \bar{\theta}_{\cdot j} = \frac{1}{P} \sum_{i=1}^{P} \theta_{ij}, \tag{34}$$

so that each skill column has zero mean across the three EPs. Centring ensures PCA captures *differential* EP–skill affinity — how each EP deviates from the average skill profile — rather than the overall level of skill occurrence.

Decomposition. The singular value decomposition (SVD) of the centred matrix $\tilde{\Theta} \in \mathbb{R}^{P \times K}$ is:

$$\tilde{\Theta} = \mathbf{U}\,\mathbf{S}\,\mathbf{V}^{\top}, \tag{35}$$

where $\mathbf{U} \in \mathbb{R}^{P \times P}$ contains the left singular vectors (EP loadings), $\mathbf{S}$ is diagonal with singular values $s_1 \geq s_2 \geq s_3 \geq 0$, and $\mathbf{V} \in \mathbb{R}^{K \times P}$ contains the right singular vectors (skill loadings). With $P = 3$ EPs and $K = 4$ skills, at most $\min(P, K) = 3$ components are available. Method C retains $r = 1$ component on parsimony grounds: the NMF rank analysis (Section 6.10) confirmed that $r = 2$ adds negligible signal ($|r(\delta_{\text{Beh}}, \bar{Y}_e)| = 0.013$, $|r(\delta_{\text{Cog}}, \bar{Y}_e)| = 0.144$).

EP scores on the first component. The score of each EP on the first principal component is:

$$\phi_i^{(1)} = s_1\, u_{i1}, \quad i = 1, 2, 3, \tag{36}$$

where u_{i1} is the i-th element of the first left singular vector. $\phi_i^{(1)}$ measures how strongly EP_i is associated with the dominant variance direction. Because Constructivism has

substantially higher conditional probabilities than Behaviourism and Cognitivism ($\theta_{3j} \gg \theta_{1j} \approx \theta_{2j}$ for all j), the first component is dominated by the Constructivism–other contrast: $\phi^{(1)}_{\text{Con}} \gg \phi^{(1)}_{\text{Beh}} \approx \phi^{(1)}_{\text{Cog}}$.

Line-level score. The Method C score for line n is the projection of the line's EP activation vector onto the first principal component, normalised by p_n:

$$\text{EPSPI}^{(C)}_n = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni}\, \phi^{(1)}_i, \qquad p_n > 0, \tag{37}$$

where $p_n = \sum_i X_{ni}$ is the number of active EPs. For a line with only Constructivism active ($p_n = 1$, $X_{n3} = 1$): $\text{EPSPI}^{(C)}_n = \phi^{(1)}_{\text{Con}}$ — a large positive value. For Behaviourism-only or Cognitivism-only lines, $\text{EPSPI}^{(C)}_n \approx 0$ after centring. When $p_n = 0$, the fallback $\text{EPSPI}^{(C)}_n = 0$ (the centred mean) is used.

Episode-level aggregation.

$$\text{EPSPI}^{(C)}_e = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \text{EPSPI}^{(C)}_n, \tag{38}$$

where n_e is the actual number of lines in episode e. Because 47 of 49 episodes have $\delta^e_{\text{Con}} \approx 0$, $\text{EPSPI}^{(C)}_e$ clusters near zero for almost all episodes, with large positive values only for Episodes 13 and 49 — the high-leverage Constructivist episodes identified in Section **??**. This compression is the direct cause of the negative Exp.Var(raw) for Method C: raw predictions cluster near zero while $\bar{Y}_e \approx 0.20$, producing systematic positive residuals $\bar{Y}_e - \text{EPSPI}^{(C)}_e > 0$ for almost every episode.

Why Exp.Var(cal) is positive despite negative Exp.Var(raw). The OLS calibration $\hat{Y}_e = a + b \cdot \text{EPSPI}^{(C)}_e$ with fitted slope $b \approx 3.2$ expands the compressed PCA scores back to the range of $\bar{Y}_e$, and intercept $a > 0$ shifts them upward. After calibration the *rank ordering* of episodes is preserved — Method C correctly identifies that high-Constructivism episodes have higher skill rates — and Exp.Var(cal) ≈ 0.49 reflects this preserved ordering. The gap Exp.Var(cal) − Exp.Var(raw) directly quantifies the scale bias from PCA compression: a method with zero scale bias satisfies Exp.Var(cal) = Exp.Var(raw).

Relationship to θ_{ij} and Methods A, B. Method C is a **rank-1 approximation** to Θ:

$$\hat{\Theta}^{(1)} = s_1\, \mathbf{u}_1\, \mathbf{v}_1^{\top}, \tag{39}$$

replacing the full 3×4 Θ matrix with its best rank-1 reconstruction. Method A uses the full Θ directly; Method C discards the second and third components. The CV r gap between

Methods A and C (0.409 vs 0.281) quantifies how much multi-dimensional structure in Θ is lost by the rank-1 constraint — specifically, the Behaviourism vs Cognitivism contrast (second component) carries additional predictive signal that Method A uses but Method C discards.

What "Train Pearson r" means for Method C. Method C has **1 free parameter** — the first principal component direction $\mathbf{u}_1$, estimated from Θ which was itself computed from all N lines. In cross-validation, Θ is recomputed from 39 training episodes, the SVD is re-run to obtain new $\phi_i^{(1)}$ values, and $\text{EPSPI}_e^{(C)}$ is computed for the 10 test episodes using these training-derived component scores. The larger Train-to-CV drop for Method C compared to Methods A and B reflects additional estimation variance from the SVD decomposition of a 3×4 matrix estimated from only 39 episodes.

Properties. Method C has **1 free parameter** and is the first method on the Dimensionality Reduction axis. Its key advantage is robustness to correlated EP–skill columns: if two skills S_j and $S_{j'}$ are highly correlated across EPs, PCA consolidates them into a single component rather than treating them as independent signals. Its key limitation is scale compression caused by the Constructivism dominance of the first component — Behaviourism and Cognitivism lines receive near-zero scores regardless of their EP flags, making raw predictions uninformative for the 98.9% of lines with no Constructivism activation. Method H (NMF) addresses this by imposing non-negativity, preventing Behaviourism and Cognitivism from being suppressed in the component scores, while Method D (Correspondence Analysis) works in the chi-squared metric of the EP–skill contingency table rather than the Euclidean metric of Θ, giving a more balanced weighting to rare EPs.

6.9 Method D: Correspondence Analysis

Method C applies PCA to the conditional probability matrix Θ in the Euclidean metric, treating each cell θ_{ij} as equally important. This gives disproportionate influence to Constructivism — whose θ_{3j} values are far from the row mean — while suppressing the subtler Behaviourism–Cognitivism contrast. Method D addresses this through **Correspondence Analysis** (CA), which decomposes the EP–skill *contingency table* in the **chi-squared metric**, weighting each cell by its deviation from statistical independence relative to its expected frequency [12**?**]. CA is the natural multivariate method for binary co-occurrence data: it simultaneously finds the low-dimensional representation of EPs and skills that best recovers the association structure of the contingency table, giving proportionally more weight to rare EP–skill combinations than PCA does.

Input matrix: the contingency table. CA operates on the raw co-occurrence count matrix $\mathbf{N} \in \mathbb{Z}^{P\times K}$, where:

$$N_{ij} = \sum_{n=1}^{N} X_{ni}\, Y_{nj}, \tag{40}$$

is the number of lines in which EP_i and S_j co-occur simultaneously — the unnormalised numerator of $P(\mathrm{EP}_i \cap S_j)$ from Equation (**??**). Unlike PCA, CA does not require column-centring because the chi-squared metric already centres the decomposition around the independence model.

Chi-squared decomposition. Let $r_i = N_{i\cdot}/N_{\cdot\cdot}$ be the row mass (marginal EP proportion) and $c_j = N_{\cdot j}/N_{\cdot\cdot}$ be the column mass (marginal skill proportion), where $N_{\cdot\cdot} = \sum_{i,j} N_{ij}$. Note that $r_i = P(\mathrm{EP}_i)$ and $c_j = P(S_j)$ from Equation (**??**). CA decomposes the matrix of standardised residuals from independence:

$$s_{ij} = \frac{p_{ij} - r_i c_j}{\sqrt{r_i c_j}}, \qquad p_{ij} = \frac{N_{ij}}{N_{\cdot\cdot}} = P(\mathrm{EP}_i \cap S_j), \tag{41}$$

via SVD: $\mathbf{S} = \mathbf{U}\,\mathbf{D}\,\mathbf{V}^\top$, where s_{ij} is the (i,j) element of $\mathbf{S}$ and $d_1 \geq d_2 \geq d_3$ are the singular values. The total inertia $\sum_k d_k^2$ equals the normalised chi-squared statistic $\chi^2/N_{\cdot\cdot}$, measuring total EP–skill association [12].

EP coordinates on the first axis. The standard row coordinates of each EP on the first CA axis are:

$$\psi_i^{(1)} = \frac{u_{i1}\, d_1}{\sqrt{r_i}}, \quad i = 1, 2, 3, \tag{42}$$

where u_{i1} is the i-th element of the first left singular vector. The $1/\sqrt{r_i}$ scaling gives proportionally larger coordinates to rare EPs: Constructivism ($r_{\mathrm{Con}} = 0.011$) receives a much larger coordinate than in PCA, and Behaviourism and Cognitivism are meaningfully separated because their chi-squared residuals s_{ij} differ non-trivially even though their absolute θ_{ij} values are similar.

Sign normalisation. CA coordinates have an arbitrary sign: multiplying all row and column coordinates by -1 produces an equivalent solution. The sign is anchored so that $\psi_{\mathrm{Con}}^{(1)} > 0$, placing Constructivism on the positive side of the first axis consistent with its high skill-conducive profile.

Rescaling to $[0,1]$. After sign normalisation, $\psi_i^{(1)}$ may be negative for Behaviourism or Cognitivism, producing negative EPSPI scores. To preserve the interpretation of EPSPI

as a skill potential in $[0, 1]$, coordinates are linearly rescaled:

$$\tilde{\psi}_i^{(1)} = \frac{\psi_i^{(1)} - \min_{i'} \psi_{i'}^{(1)}}{\max_{i'} \psi_{i'}^{(1)} - \min_{i'} \psi_{i'}^{(1)}}, \tag{43}$$

which preserves the rank ordering of EPs on the first axis.

Line-level score. The Method D score for line n is:

$$\mathrm{EPSPI}_n^{(D)} = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni}\, \tilde{\psi}_i^{(1)}, \qquad p_n > 0, \tag{44}$$

where $p_n = \sum_i X_{ni}$. For a line with only Constructivism active: $\mathrm{EPSPI}_n^{(D)} = \tilde{\psi}_{\mathrm{Con}}^{(1)} = 1.0$ (after normalisation). For a line with only Behaviourism active: $\mathrm{EPSPI}_n^{(D)} = \tilde{\psi}_{\mathrm{Beh}}^{(1)}$, which under the chi-squared metric reflects the *relative* deviation of Behaviourism's skill profile from independence — more informative than PCA's Euclidean distance from the row mean. When $p_n = 0$, the fallback $\mathrm{EPSPI}_n^{(D)} = 0.5$ (midpoint of $[0, 1]$) is used.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(D)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(D)}, \tag{45}$$

where n_e is the actual number of lines in episode e. Unlike Method C, where almost all episodes received near-zero scores, Method D's $1/\sqrt{r_i}$ scaling amplifies the Behaviourism–Cognitivism contrast, giving meaningfully different $\mathrm{EPSPI}_e^{(D)}$ values across episodes even when Constructivism is absent. This avoids the scale compression that caused Method C's negative Exp.Var(raw).

Relationship to Methods A and C. CA and PCA both decompose a matrix derived from Θ but in different metrics:

- **PCA (Method C)** decomposes $\tilde{\Theta}$ in the Euclidean metric — all cells θ_{ij} equally weighted. Constructivism dominates because its θ_{3j} values are far from the column mean in absolute terms.
- **CA (Method D)** decomposes $\mathbf{S}$ (standardised residuals from independence) in the chi-squared metric — rare EP–skill combinations that deviate from independence are upweighted. Constructivism still has large coordinates, but Behaviourism and Cognitivism are also meaningfully separated through their relative deviations s_{ij}.

CA theory predicts that it finds the representation maximising the correlation between EP row scores and skill column scores simultaneously [12] — exactly the structure that

the logistic AIC rewards. This explains why Method D achieves the **best AIC** (44,358) of all eleven methods: CA directly optimises chi-squared EP–skill co-occurrence at the line level, which is precisely what the logistic likelihood rewards.

The AIC–CV tension. Method D achieves the best AIC but only CV $r = 0.296$ — among the weakest episode-level predictive validity. This tension is a genuine scientific finding, not an artefact. AIC measures **line-level discriminative power**: how well $\text{EPSPI}_n^{(D)}$ separates lines with high vs low skill rates. CV r measures **episode-level predictive validity**: how well $\text{EPSPI}_e^{(D)}$ predicts the aggregate skill rate of unseen episodes. CA optimises the former by construction. But episode-level aggregation (Layer 2) averages out much of this line-level discrimination: an episode with a balanced EP mix will have a moderate $\text{EPSPI}_e^{(D)}$ that loses the within-episode variation that drives the AIC advantage. This AIC–CV tension demonstrates that line-level discriminative power and episode-level predictive validity are genuinely distinct properties — a method can excel at one while underperforming at the other.

What "Train Pearson r" means for Method D. Method D has $P = 3$ **free parameters** — the three EP row coordinates $\tilde{\psi}_i^{(1)}$, estimated from the contingency table $\mathbf{N}$ computed from all N lines. In cross-validation, $\mathbf{N}$ is recomputed from 39 training episodes, the CA decomposition is re-run with sign normalisation and $[0, 1]$ rescaling re-applied, and $\text{EPSPI}_e^{(D)}$ is computed for the 10 test episodes using these training-derived coordinates. The in-sample Train Pearson r is inflated relative to CV r because the CA coordinates are optimised on the same 49 episodes used for evaluation.

Properties. Method D has $P = 3$ **free parameters**. It is the only method whose estimation criterion (chi-squared association) directly corresponds to the logistic AIC used for evaluation, explaining its AIC dominance. Its limitation for episode-level prediction is that the chi-squared metric upweights rare EP–skill combinations that are informative at the line level but average out at the episode level. The sign ambiguity (Equation (43)) and the need for $[0, 1]$ rescaling are minor implementation costs — the substantive CA output is the rank ordering of EPs on the first axis, which is preserved by Equation (43). Together with Methods C and H, Method D completes a systematic comparison of the rank-1 constraint on Θ under three geometric frameworks: Euclidean (PCA), chi-squared (CA), and non-negative (NMF) — examining whether the choice of metric materially affects episode-level predictive validity.

6.10 Method H: Non-Negative Matrix Factorisation (NMF)

Methods C and D apply unconstrained (PCA) and chi-squared (CA) decompositions to Θ, both of which allow negative component values. Negative values are mathematically valid

but pedagogically uninterpretable: a negative loading on Constructivism would imply that Constructivist teaching *reduces* the skill-potential score, which has no meaningful interpretation in the context of educational practice. Method H imposes the **non-negativity constraint** on both the EP factor scores and the skill factor scores, ensuring that the decomposition represents additive combinations of teaching-mode profiles — each component contributes positively to the overall EPSPI score [21**?**].

Input matrix and constraint. NMF is applied directly to the global Θ matrix (not column-centred, unlike PCA):

$$\Theta \approx \mathbf{W}\,\mathbf{H}, \qquad \mathbf{W} \geq 0, \quad \mathbf{H} \geq 0, \tag{46}$$

where $\mathbf{W} \in \mathbb{R}_{\geq 0}^{P \times r}$ contains the EP factor scores (how strongly each EP loads on each latent teaching mode) and $\mathbf{H} \in \mathbb{R}_{\geq 0}^{r \times K}$ contains the skill factor scores (how strongly each latent teaching mode is associated with each skill). The rank r controls the number of latent teaching modes extracted. Both $\mathbf{W}$ and $\mathbf{H}$ are non-negative by constraint, so every element of the approximation $\hat{\Theta} = \mathbf{WH}$ is a *non-negative additive combination* of the r latent profiles. No column-centring is applied because non-negativity and centring are incompatible: centring introduces negative values that the non-negativity constraint would then suppress, distorting the factorisation.

Why $r = 1$ component is used. With $P = 3$ EPs and $K = 4$ skills, the 3×4 matrix Θ has rank at most $\min(P, K) = 3$. A rank comparison across $r = 1, 2, 3$ was conducted by computing CV r for each rank:

Rank r	Description	CV r
1	Single Constructivism-dominated mode	0.348
2	+ Beh vs Cog contrast axis	$\lvert r(\delta_{\text{Beh}}, \bar{Y}_e)\rvert = 0.013$, $\lvert r(\delta_{\text{Cog}}, \bar{Y}_e)\rvert = 0.144$ (negligib
3	Perfect reconstruction of Θ	Equivalent to Method A

At $r = 2$, the second component captures the Behaviourism–Cognitivism contrast, but this axis carries negligible signal in predicting $\bar{Y}_e$ ($|r| < 0.15$ for both EPs on this axis). At $r = 3$, NMF perfectly reconstructs Θ (rank-3 factorisation of a rank-3 matrix), making Method H equivalent to Method A — no compression is achieved. Method H therefore uses $r = 1$ on parsimony grounds: the dominant Constructivism mode captures the only predictively relevant latent structure in Θ [21].

Estimation. For $r = 1$, $\mathbf{W} \in \mathbb{R}^{3\times 1}_{\geq 0}$ and $\mathbf{H} \in \mathbb{R}^{1\times 4}_{\geq 0}$, so the factorisation reduces to a rank-1 outer product:

$$\Theta \approx \mathbf{w}\,\mathbf{h}^\top, \qquad \mathbf{w} = [w_1, w_2, w_3]^\top \geq 0, \quad \mathbf{h} = [h_1, h_2, h_3, h_4]^\top \geq 0, \tag{47}$$

where $w_i \geq 0$ is the loading of EP_i on the single latent teaching mode and $h_j \geq 0$ is the loading of skill S_j on that mode. Parameters are estimated by minimising the Frobenius-norm reconstruction error:

$$\min_{\mathbf{w}\geq 0,\, \mathbf{h}\geq 0} \left\|\Theta - \mathbf{w}\,\mathbf{h}^\top\right\|_F^2 = \min_{\mathbf{w},\,\mathbf{h}} \sum_{i=1}^{P}\sum_{j=1}^{K} \left(\theta_{ij} - w_i h_j\right)^2, \tag{48}$$

solved by the multiplicative update rules of **?**], which guarantee $\mathbf{w} \geq 0$ and $\mathbf{h} \geq 0$ are maintained throughout optimisation. The observed fitted values for the corpus are:

$$\mathbf{w} = [0.014,\ 0.000,\ 1.000]^\top \quad \text{(Beh, Cog, Con)},$$

$$\mathbf{h} = [0.779,\ 0.038,\ 0.748,\ 0.346] \quad \text{(PS, Cr, CL, CT)},$$

with reconstruction MSE $= 0.394$. The near-zero $w_{\mathrm{Cog}} = 0.000$ confirms that the single latent mode is almost entirely Constructivism: NMF has identified that Constructivism is the only EP with a substantially different skill profile from the global baseline, and Behaviourism and Cognitivism are effectively indistinguishable at rank 1.

Interpretation of the latent teaching mode. The rank-1 factorisation $\hat{\theta}_{ij} = w_i h_j$ decomposes the EP–skill affinity into two independent factors:

- w_i: how strongly EP_i activates the latent teaching mode. Constructivism ($w_3 = 1.000$) fully activates it; Behaviourism ($w_1 = 0.014$) barely activates it; Cognitivism ($w_2 = 0.000$) does not activate it at all.
- h_j: how strongly the latent mode is associated with skill S_j. Problem Solving ($h_1 = 0.779$) and Collaborative Learning ($h_3 = 0.748$) are most strongly associated; Creativity ($h_2 = 0.038$) is weakest.

The latent teaching mode can therefore be interpreted as a **Constructivist–collaborative skill mode**: it is almost exclusively activated by Constructivism, and it primarily predicts Problem Solving and Collaborative Learning. This aligns with established educational theory: Constructivist pedagogy emphasises active knowledge construction through collaborative problem-solving [**?** **?**].

Line-level score. The Method H score for line n is the NMF-reconstructed conditional skill probability, normalised by p_n:

$$\mathrm{EPSPI}_n^{(H)} = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni} \left(\frac{1}{K} \sum_{j=1}^{K} w_i h_j \right) = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni}\, w_i \cdot \bar{h}, \quad p_n > 0, \tag{49}$$

where $\bar{h} = \frac{1}{K} \sum_{j=1}^{K} h_j$ is the mean skill loading of the latent mode. The inner average $\frac{1}{K} \sum_j w_i h_j = w_i \bar{h}$ factorises into the EP loading w_i and the constant $\bar{h}$, so Method H is equivalent to Method A with θ_{ij} replaced by the rank-1 approximation $\hat{\theta}_{ij} = w_i h_j$.

Example: line with only Constructivism active. Setting $X_{n3} = 1$, $p_n = 1$:

$$\mathrm{EPSPI}_n^{(H)} = w_3 \cdot \bar{h} = 1.000 \times \tfrac{1}{4}(0.779 + 0.038 + 0.748 + 0.346) = 1.000 \times 0.478 = 0.478.$$

Compare to Method A: $\mathrm{EPSPI}_n^{(A)} = 0.510$. The difference (0.032) is the reconstruction error for the Constructivism row — NMF slightly underestimates the Constructivism affinity because it must find a single $\mathbf{w}\mathbf{h}^\top$ that approximates all three EP rows simultaneously under non-negativity.

Example: line with only Cognitivism active. Setting $X_{n2} = 1$, $p_n = 1$:

$$\mathrm{EPSPI}_n^{(H)} = w_2 \cdot \bar{h} = 0.000 \times 0.478 = 0.000.$$

A pure-Cognitivism line receives EPSPI = 0, indicating that Cognitivism contributes nothing to the latent Constructivist–collaborative mode. Compare to Method A: $\mathrm{EPSPI}_n^{(A)} = 0.147$. This is the scale compression that causes $\mathrm{EPSPI}_e^{(H)} \approx 0.028$ to be far below $\bar{Y}_e \approx 0.20$ for most episodes, and is the direct cause of the negative Exp.Var(raw) for Method H. When $p_n = 0$, the fallback $\mathrm{EPSPI}_n^{(H)} = \bar{h}/P = 0.478/3 = 0.159$ is used.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(H)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(H)}, \tag{50}$$

where n_e is the actual number of lines in episode e. Because $w_{\mathrm{Cog}} = 0.000$ and $w_{\mathrm{Beh}} = 0.014 \approx 0$, episodes with no Constructivism lines receive $\mathrm{EPSPI}_e^{(H)} \approx 0$ regardless of their Behaviourism or Cognitivism content. This is the same scale compression as Method C, but for a different reason: in PCA, compression arises from Constructivism dominating the principal component direction; in NMF, it arises from Behaviourism and Cognitivism having near-zero w_i values because the non-negativity constraint prevents them from having large positive loadings on a Constructivism-dominated mode.

Relationship to Methods A, C, and D. All three dimensionality reduction methods (C, D, H) are rank-1 approximations to Θ, differing only in metric and constraint:

Method	Input	Metric	Constraint
C (PCA)	$\tilde{\Theta}$ (centred)	Euclidean	None
D (CA)	$\mathbf{N}$ (counts)	Chi-squared	None
H (NMF)	Θ (uncentred)	Euclidean	$\mathbf{W}, \mathbf{H} \geq 0$

All three produce similar CV r values (0.281–0.348), confirming that the rank-1 constraint is the binding limitation, not the choice of metric or non-negativity. The additional interpretability of NMF — additive teaching-mode profiles with non-negative loadings — comes at no cost in predictive performance relative to PCA or CA. Method A (full Θ, no rank constraint) outperforms all three (CV $r = 0.409$), quantifying the predictive information contained in the rank-2 and rank-3 components of Θ that all dimensionality reduction methods discard.

What "Train Pearson r" means for Method H. Method H has $P + K = 7$ **free parameters** (w_1, w_2, w_3 and h_1, h_2, h_3, h_4), estimated by minimising the Frobenius-norm reconstruction error on the global Θ matrix. Because Θ is computed from all N lines, the in-sample Train Pearson r reflects in-sample association. In cross-validation, Θ is recomputed from 39 training episodes, NMF is re-run to obtain new $\mathbf{w}$ and $\mathbf{h}$ vectors, and $\text{EPSPI}_e^{(H)}$ is computed for the 10 test episodes using these training-derived factors. The drop from Train $r \approx 0.70$ to CV $r \approx 0.35$ is moderate — comparable to Methods A and B — reflecting the low estimation variance of a $P + K = 7$ parameter model fitted to a $3 \times 4 = 12$-cell matrix.

Properties. Method H has $P + K = 7$ **free parameters** — more than Methods C and D (1 and 3 respectively) but far fewer than the parametric methods E, F, I (16–20). Its key advantage over Methods C and D is **interpretability**: the non-negativity constraint ensures that w_i and h_j are genuine part-based representations of EP–skill structure [21], supporting the pedagogical interpretation of the latent mode as "Constructivist–collaborative skill activation." Its key limitation is the same scale compression as Method C: because $w_{\text{Beh}} \approx w_{\text{Cog}} \approx 0$, the 98.9% of lines without Constructivism activation receive near-zero scores, producing negative Exp.Var(raw) for all episodes except 13 and 49. Together, Methods C, D, and H demonstrate that the rank-1 constraint on Θ — regardless of metric or non-negativity — captures only the Constructivism–other contrast, and that predicting skill deployment in predominantly Behaviourist or Cognitivist episodes requires the full Θ matrix used in Method A or the parametric modelling of Methods E, G, and I.

6.11 Method F: Naïve Bayes

Methods A through E all estimate $P(S_j \mid \mathrm{EP}_i)$ — the probability of a skill given an EP — either by direct frequency counting or by a discriminative log-odds model. Method F reverses this direction: it models $P(X_{n\cdot} \mid S_j)$ — the probability of observing a particular EP profile given that a skill is present or absent — and then applies Bayes' theorem to recover $P(S_j \mid X_{n\cdot})$. This **generative** approach is the classical Naïve Bayes classifier [**? ?**], which assumes that the P EP flags are **conditionally independent** given the skill label $Y_{nj} \in \{0, 1\}$. Method F is the first method on the Predictive Modelling axis: it treats EPSPI_n as a regression target rather than as a direct estimate of θ_{ij}, using the EP flags $X_{n\cdot}$ as features to predict Y_{nj} and then averaging the predictions across skills.

Model specification. For each skill S_j, the Naïve Bayes model specifies:

1. **Class prior**: $\pi_j = P(Y_{nj} = 1)$, the marginal probability that skill S_j is observed in a randomly chosen line. This is estimated as:

$$\hat{\pi}_j = \frac{\sum_{n=1}^{N} Y_{nj} + \alpha}{N + 2\alpha}, \tag{51}$$

   using Laplace smoothing with $\alpha = 1$.

2. **Conditional likelihood**: for each EP i and skill label $y \in \{0, 1\}$, the probability that EP_i is active given skill label y:

$$\hat{\lambda}_{ij}^{(y)} = P(X_{ni} = 1 \mid Y_{nj} = y) = \frac{\sum_n X_{ni}\, \mathbf{1}[Y_{nj} = y] + \alpha}{\sum_n \mathbf{1}[Y_{nj} = y] + 2\alpha}, \tag{52}$$

   again with Laplace smoothing $\alpha = 1$.

The model has $(2P + 1) \times K = 7 \times 4 = 28$ free parameters: for each of the $K = 4$ skills, one prior $\hat{\pi}_j$, $P = 3$ skill-present likelihoods $\hat{\lambda}_{ij}^{(1)}$, and $P = 3$ skill-absent likelihoods $\hat{\lambda}_{ij}^{(0)}$.

Conditional independence assumption. The defining assumption of Naïve Bayes is that the P EP flags are mutually independent conditional on the skill label:

$$P(X_{n\cdot} \mid Y_{nj} = y) = \prod_{i=1}^{P} P(X_{ni} \mid Y_{nj} = y) = \prod_{i=1}^{P} \left(\hat{\lambda}_{ij}^{(y)}\right)^{X_{ni}} \left(1 - \hat{\lambda}_{ij}^{(y)}\right)^{1 - X_{ni}}. \tag{53}$$

This assumption is demonstrably violated in this corpus: Behaviourism and Cognitivism co-occur more often than independence would predict (δ_{Beh} and δ_{Cog} are positively correlated across episodes), so $P(X_{n,\mathrm{Beh}} = 1, X_{n,\mathrm{Cog}} = 1 \mid Y_{nj})$ is not separable into independent factors. Despite this, Naïve Bayes is known to produce accurate probability estimates under EP correlation when the classification boundary is approximately linear in the

feature space [?], which is the case here given that $P = 3$ binary features define at most $2^3 = 8$ distinct EP profiles.

Posterior probability via Bayes' theorem. Applying Bayes' theorem with the conditional independence assumption, the posterior probability that skill S_j is present for line n is:

$$\hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) = \frac{\hat{\pi}_j \prod_{i=1}^{P} \left(\hat{\lambda}_{ij}^{(1)}\right)^{X_{ni}} \left(1 - \hat{\lambda}_{ij}^{(1)}\right)^{1-X_{ni}}}{\sum_{y\in\{0,1\}} (1-\hat{\pi}_j)^{1-y}\, \hat{\pi}_j^{y} \prod_{i=1}^{P} \left(\hat{\lambda}_{ij}^{(y)}\right)^{X_{ni}} \left(1 - \hat{\lambda}_{ij}^{(y)}\right)^{1-X_{ni}}}. \tag{54}$$

In practice, Equation (54) is computed in log-space to avoid numerical underflow:

$$\log \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) \propto \log \hat{\pi}_j + \sum_{i=1}^{P} \Big[X_{ni} \log \hat{\lambda}_{ij}^{(1)} + (1 - X_{ni}) \log(1 - \hat{\lambda}_{ij}^{(1)}) \Big], \tag{55}$$

with normalisation applied after exponentiation.

Relationship to θ_{ij} and Method E. Method F and Method E both estimate $\hat{P}(Y_{nj} = 1 \mid X_{n\cdot})$ and average over skills to produce EPSPI_n. They differ in *how* this conditional probability is estimated:

- **Method E (Logistic Regression)** is *discriminative*: it directly models logit $P(Y_{nj} = 1 \mid X_{n\cdot})$ as a linear function of $X_{n\cdot}$, optimising parameters by maximising the conditional log-likelihood $\sum_n \log P(Y_{nj} \mid X_{n\cdot}; \hat{\beta})$.
- **Method F (Naïve Bayes)** is *generative*: it models $P(X_{n\cdot} \mid Y_{nj})$ and $P(Y_{nj})$ separately, then recovers $P(Y_{nj} \mid X_{n\cdot})$ via Bayes' theorem. The joint log-likelihood $\sum_n \log P(X_{n\cdot}, Y_{nj}; \hat{\lambda}, \hat{\pi})$ is optimised, which uses more information than the conditional log-likelihood.

When the conditional independence assumption holds, Naïve Bayes is statistically equivalent to logistic regression [?]: the Naïve Bayes posterior (54) is a logistic function of a linear combination of the X_{ni} flags, with $\log(\hat{\lambda}_{ij}^{(1)}/\hat{\lambda}_{ij}^{(0)})$ playing the role of the logistic regression coefficient $\hat{\beta}_{ij}$. The empirical differences in CV r and AIC between Methods E and F therefore measure the cost of the conditional independence assumption in this corpus.

Line-level score. The Method F score for line n is the mean posterior skill probability across all K skills:

$$\text{EPSPI}_n^{(F)} = \frac{1}{K} \sum_{j=1}^{K} \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}), \tag{56}$$

where $\hat{P}(Y_{nj} = 1 \mid X_{n\cdot})$ is computed via Equation (54). There is no p_n normalisation — all three EP flags enter the likelihood product (53) simultaneously, with the Laplace-smoothed likelihoods $\hat{\lambda}_{ij}^{(y)}$ absorbing the contribution of each active EP.

Example: line with only Constructivism active. Setting $X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$:

$$\log \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) \propto \log \hat{\pi}_j + \log(1 - \hat{\lambda}_{1j}^{(1)}) + \log(1 - \hat{\lambda}_{2j}^{(1)}) + \log \hat{\lambda}_{3j}^{(1)}.$$

For Problem Solving ($j = 1$), where Constructivism is strongly associated with skill presence ($\hat{\lambda}_{3,\mathrm{PS}}^{(1)}$ is high), the posterior $\hat{P}(Y_{n,\mathrm{PS}} = 1 \mid X_{n\cdot})$ is large, contributing a high value to the mean in Equation (56).

Example: line with Behaviourism and Cognitivism active. Setting $X_{n1} = 1$, $X_{n2} = 1$, $X_{n3} = 0$:

$$\log \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) \propto \log \hat{\pi}_j + \log \hat{\lambda}_{1j}^{(1)} + \log \hat{\lambda}_{2j}^{(1)} + \log(1 - \hat{\lambda}_{3j}^{(1)}).$$

Under the conditional independence assumption, the contributions of Behaviourism and Cognitivism multiply in probability space — they reinforce each other if both are positively associated with skill presence, or partially cancel if their associations conflict. Method E handles the same co-occurrence by additive log-odds; Method F handles it by multiplicative probabilities — the two approaches agree when the independence assumption holds.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(F)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(F)}, \tag{57}$$

where n_e is the actual number of lines in episode e. As with Methods E and I, $\mathrm{EPSPI}_n^{(F)}$ varies line by line through the EP profile $[X_{n1}, X_{n2}, X_{n3}]$, so the episode mean is a genuine average of heterogeneous line scores.

Why global estimation and not episode-conditional. As with Methods E and I, $\hat{\pi}_j$ and $\hat{\lambda}_{ij}^{(y)}$ are estimated from all $N = 19{,}181$ lines jointly — not episode by episode. Fitting Naïve Bayes per episode would use the same episode's Y_{nj} observations to estimate the likelihoods and to compute $\bar{Y}_e$, creating a direct within-episode circular dependency between predictor and outcome. In cross-validation, all parameters are re-estimated from the 39 training episodes and applied to 10 test episodes where Y_{nj} was never seen during fitting. Constructivism's sparsity (1.1% of lines) is handled naturally by the Laplace smoothing in Equations (51) and (52) — unlike Methods E and G, no convergence guard or

coefficient bound is required. This is a practical advantage of the generative approach: the likelihood parameters $\hat{\lambda}_{ij}^{(y)}$ are simple frequency ratios with guaranteed finite values under Laplace smoothing, whereas logistic regression coefficients diverge under quasi-complete separation.

What "Train Pearson r" means for Method F. Method F has $(2P+1) \times K = 28$ **free parameters** estimated from all N lines. The in-sample Train Pearson r reflects genuine in-sample fit under the generative likelihood, with the same episode-level caveat as Methods E and I: parameters are estimated from all 49 episodes' line-level observations, so the Train r is not a held-out evaluation. Method F achieves CV $r = 0.340$ and CV MSE $= 0.008730$ — moderately lower than Method I (CV $r = 0.450$) but competitive with Methods A and J. The gap between Method F and Method E in CV r measures the empirical cost of the conditional independence assumption in this corpus under cross-validation.

Properties. Method F has $(2P+1) \times K = 28$ **free parameters** — more than Method E (16) because the generative model estimates both $P(Y_{nj})$ and $P(X_{ni} \mid Y_{nj})$ separately, requiring twice as many EP-related parameters as the discriminative model. Its key advantage over Method E is robustness to quasi-complete separation: Laplace-smoothed frequency counts never diverge, whereas logistic regression coefficients for rare EPs (Constructivism) require an explicit convergence guard. Its key limitation is the conditional independence assumption (Equation (53)), which is violated in this corpus — Behaviourism and Cognitivism co-occur more frequently than independence would predict. In the generative–discriminative comparison of **?**], discriminative models (Method E) tend to outperform generative ones (Method F) in the large-N regime because they optimise the correct conditional criterion; generative models may outperform in the small-N regime due to faster parameter convergence. With $N = 19{,}181$ lines and $E = 49$ episodes, this corpus is large enough that Method E is expected to outperform Method F when the independence assumption is violated — which the CV results confirm.

6.12 Method J: Random Forest

Methods E (Logistic Regression) and F (Naïve Bayes) both impose a parametric functional form on $P(Y_{nj} \mid X_{n\cdot})$ — log-odds linearity and conditional independence respectively. Method J replaces these assumptions with a **non-parametric ensemble** of decision trees, each fitted to a bootstrap sample of the training data with a random subset of features considered at each split [5]. Random Forest makes no assumption about the form of the EP–skill decision boundary and can therefore capture non-linear interactions between EP flags and episode-conditional skill probabilities that the parametric methods cannot

represent. Method J is the second method on the Predictive Modelling axis and serves as the primary non-parametric benchmark for Methods E, F, and K.

Feature construction. A naïve Random Forest using only the $P = 3$ binary EP flags $[X_{n1}, X_{n2}, X_{n3}]$ as features would face a severe representation problem: with $P = 3$ binary inputs, only $2^P = 8$ distinct feature vectors exist across all $N = 19{,}181$ lines. Every tree in the ensemble would see at most 8 unique splitting configurations, making the ensemble equivalent to a lookup table and providing no advantage over the non-parametric θ_{ij} estimate of Method A. Method J therefore uses the same **15-feature input** as Method K (SVM), combining the EP flags with the episode-conditional skill probability estimates:

$$\mathbf{x}_n = \big[\underbrace{X_{n1},\, X_{n2},\, X_{n3}}_{\text{EP flags } (P=3)},\quad \underbrace{\hat{\theta}^{e_n}_{11},\, \hat{\theta}^{e_n}_{12},\, \ldots,\, \hat{\theta}^{e_n}_{PK}}_{\text{episode-conditional } \hat{\Theta}^{e_n} \text{ flattened } (P\times K=12)}\big] \in \mathbb{R}^{15}, \tag{58}$$

where $\hat{\theta}^{e_n}_{ij}$ is the Laplace-smoothed episode-conditional conditional probability (Equation (**??**)) for the episode containing line n. The 12 elements of the flattened $\hat{\Theta}^{e_n}$ matrix encode how the local EP–skill affinity profile of episode e_n differs from the global Θ, providing the Random Forest with information about the pedagogical context of each line beyond its own EP flags.

Leakage-free episode-conditional features. The $\hat{\theta}^{e_n}_{ij}$ features require special handling to prevent leakage. If $\hat{\theta}^{e_n}_{ij}$ is computed from all lines in episode e_n — including the line n being predicted — the feature incorporates Y_{nj} from the same line used as the outcome, which is direct leakage. The same leakage-free protocol used for Method K is applied: in cross-validation, for each test episode e, $\hat{\theta}^{e}_{ij}$ is computed from the 39 training episodes only. Test episode lines therefore use training-derived $\hat{\theta}^{e_n}_{ij}$ values that have never seen the test-episode Y_{nj} observations. At the line level within a training episode, leave-one-out (LOO) episode-conditional estimates $\hat{\theta}^{-e_n}_{ij}$ are used during tree fitting, removing line n from the episode-conditional count before computing the feature for that line. This ensures the full CV evaluation is leakage-free at both the episode and line levels.

Model specification. For each skill S_j, a Random Forest regressor is fitted with outcome $Y_{nj} \in \{0, 1\}$ and feature vector $\mathbf{x}_n \in \mathbb{R}^{15}$:

$$\hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n) = \frac{1}{T} \sum_{t=1}^{T} f_t^{(j)}(\mathbf{x}_n), \tag{59}$$

where T trees are trained and $f_t^{(j)}(\mathbf{x}_n) \in \{0, 1\}$ is the prediction of the t-th tree for line n. Each tree $f_t^{(j)}$ is grown on a bootstrap sample of the training lines, and at each node a random subset of $m = \lfloor\sqrt{15}\rfloor = 3$ features is considered for splitting, reducing the

correlation between trees and improving ensemble variance reduction [5]. Trees are grown to full depth (no pruning) because individual trees are high-variance weak learners — the ensemble averaging in Equation (59) provides regularisation. The hyperparameters T (number of trees) and m (features per split) are set to $T = 500$ and $m = \lfloor\sqrt{15}\rfloor = 3$ following the defaults of **(author?)** [5] and verified by out-of-bag (OOB) error stabilisation.

Relationship to Methods E, F, and A. Method J shares the same prediction target as Methods E and F — $\hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n)$ averaged over skills — but replaces their parametric functional forms with a non-parametric ensemble. Three specific comparisons are informative:

- **Method J vs Method E** measures the cost of log-odds linearity: if the EP–skill boundary is non-linear, Method J should outperform Method E in CV r. The observed CV r values (J: 0.394; E: 0.400) indicate that the decision boundary is approximately linear and the extra flexibility of Random Forest provides negligible benefit over logistic regression.

- **Method J vs Method A** measures the benefit of episode-conditional features: Method A uses only global θ_{ij} while Method J uses both EP flags and episode-conditional $\hat{\theta}_{ij}^{e_n}$. The similar CV r values (J: 0.394; A: 0.409) suggest that the additional episode-local information in $\hat{\theta}_{ij}^{e_n}$ does not translate to improved episode-level prediction, possibly because episode-conditional estimates are sparse and noisy for rare EPs (Constructivism).

- **Method J vs Method K** (SVM) uses the same 15 features and the same leakage-free protocol, making it a pure comparison of non-parametric tree ensemble vs kernel-based classifier on identical inputs. The CV r values (J: 0.394; K: 0.307) indicate that the ensemble averaging of Random Forest handles the sparse, unbalanced EP–skill co-occurrence structure better than the SVM margin criterion.

Line-level score. The Method J score for line n is the mean posterior skill probability across all K skills:

$$\mathrm{EPSPI}_n^{(J)} = \frac{1}{K}\sum_{j=1}^{K} \hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n), \tag{60}$$

where each $\hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n)$ is the ensemble mean from Equation (59). Unlike Methods A and B, which normalise by p_n, Method J receives all 15 features simultaneously with no manual normalisation — the tree splitting criterion implicitly handles the interaction between EP flags and episode-conditional $\hat{\theta}_{ij}^{e_n}$ features.

Example: line with only Constructivism active in a high-Constructivism episode. For a line with $X_{n3} = 1$, $X_{n1} = X_{n2} = 0$ in an episode where $\hat{\theta}_{3j}^{e_n}$ is large (e.g. Episode 13 where Constructivism dominates), the feature vector $\mathbf{x}_n$ has $X_{n3} = 1$ together with large $\hat{\theta}_{3j}^{e_n}$ values — a highly distinctive profile that trees can isolate in early splits. Contrast this with Method A, which uses only global θ_{3j} regardless of the episode: Method J can distinguish a Constructivism-active line in Episode 13 (where Constructivism is established and consistent) from one in a mixed episode where Constructivism appears sporadically, exploiting the richer feature representation.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(J)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(J)}, \tag{61}$$

where n_e is the actual number of lines in episode e. As with Methods E, F, and I, $\mathrm{EPSPI}_n^{(J)}$ varies line by line through both the EP profile and the episode-conditional features, so the episode mean is a genuine average of heterogeneous line scores.

Feature importance. Random Forest provides a natural measure of feature importance via the mean decrease in node impurity (Gini importance) across all trees and all skills [5]:

$$\mathrm{Imp}(f_k) = \frac{1}{K} \sum_{j=1}^{K} \sum_{t=1}^{T} \sum_{\text{node } v \in f_t^{(j)}} \Delta G_v \cdot \mathbf{1}[\text{feature } k \text{ used at } v], \tag{62}$$

where ΔG_v is the reduction in Gini impurity at node v. Feature importance diagnostics serve two purposes here. First, they reveal whether the 12 episode-conditional $\hat{\theta}_{ij}^{e_n}$ features contribute beyond the 3 EP flags — if importance is concentrated on the EP flags alone, the 15-feature design collapses to the 3-feature case and Method J is redundant with Method A. Second, they identify which specific $\hat{\theta}_{ij}^{e_n}$ values drive prediction — a strong $\hat{\theta}_{\mathrm{Con,CL}}^{e_n}$ importance, for example, would indicate that the within-episode Constructivism–Collaborative Learning affinity is the primary predictor of episode-level skill rates, consistent with the NMF latent mode interpretation (Section 6.10).

What "Train Pearson r" means for Method J. Method J has no closed-form parameter count — Random Forest is a non-parametric method whose complexity grows with the number of trees and tree depth. The in-sample Train Pearson r is computed by predicting $\mathrm{EPSPI}_e^{(J)}$ using the fitted ensemble on the same 49 episodes used for training, and reflects genuine in-sample fit. Random Forest is known to be approximately unbiased in training prediction (each tree is grown to full depth on a bootstrap sample, so the ensemble mean is close to the true training-set response), but the Train r is still inflated

relative to CV r because the trees were grown on data that includes the target episodes. The out-of-bag (OOB) error provides an alternative in-training estimate of generalisation error that uses only trees for which line n was not in the bootstrap sample — the OOB r is reported alongside CV r as a consistency check. Method J achieves CV $r = 0.394$ and CV MSE $= 0.009359$, placing it between Methods E and F and confirming that the non-parametric ensemble provides no advantage over logistic regression at this sample size.

Why AIC is highest for Method J. Method J achieves the highest AIC (48,508) of all eleven methods despite its competitive CV r. The AIC is computed from a fixed external logistic model using $\text{EPSPI}_n^{(J)}$ as the predictor (Equation (**??**)), so it measures how well the *ranking* of lines by $\text{EPSPI}_n^{(J)}$ separates skill-present from skill-absent lines. Random Forest predictions are smooth averages of $T = 500$ tree votes, which compresses extreme predictions toward $[0.2, 0.8]$ — the ensemble averaging that reduces variance also reduces the spread of predicted values. This compression produces a flatter logistic discrimination curve, increasing AIC relative to methods (E, F, I) that produce more extreme predictions for high-Constructivism lines. The AIC–CV tension for Method J is therefore opposite in nature to that of Method D: Method D has excellent AIC but weak CV r (line-level discrimination without episode-level prediction), while Method J has weak AIC but competitive CV r (episode-level prediction without extreme line-level discrimination).

Properties. Method J is a non-parametric ensemble with no closed-form parameter count. Its key advantage is freedom from parametric assumptions: it can capture non-linear EP interactions, threshold effects, and feature interactions between the EP flags and episode-conditional $\hat{\theta}_{ij}^{e_n}$ values that logistic regression cannot represent. Its key limitation in this corpus is that the EP–skill decision boundary is approximately linear — the comparative results confirm that parametric methods E and I achieve higher CV r with fewer parameters and without the computational overhead of growing $T = 500$ trees per skill. The feature importance output (Equation (62)) provides the primary scientific contribution of Method J beyond its predictive performance: it quantifies the relative importance of global EP structure (flags) vs local episode context ($\hat{\theta}_{ij}^{e_n}$) in predicting skill deployment, which is not directly accessible from the coefficient estimates of Methods E or I. Method K (SVM) provides a complementary non-parametric comparison using the same 15-feature input with a kernel-based margin criterion rather than an ensemble criterion.

6.13 Method J: Random Forest

Methods E (Logistic Regression) and F (Naïve Bayes) both impose a parametric functional form on $P(Y_{nj} \mid X_{n\cdot})$ — log-odds linearity and conditional independence respectively. Method J replaces these parametric assumptions with a **non-parametric ensemble** of

decision trees [5], fitted to bootstrap samples of the training data using only the $P = 3$ binary EP flags $[X_{n1}, X_{n2}, X_{n3}]$ as features. Method J serves as the primary non-parametric benchmark for Methods E, F, and K on the Predictive Modelling axis.

The lookup-table reduction. With $P = 3$ binary EP flags, there are exactly $2^P = 2^3 = 8$ distinct feature vectors $[X_{n1}, X_{n2}, X_{n3}] \in \{0, 1\}^3$. Every line sharing the same EP pattern necessarily receives the same Random Forest prediction, because the ensemble can only split on the three binary inputs and all $2^3 = 8$ terminal nodes are fully identified by the pattern. The ensemble (n_estimators = 200, min_samples_leaf = 10) therefore reduces exactly to a **lookup table**: a $2^P \times K = 8 \times 4$ matrix of predicted skill probabilities, one row per EP pattern and one column per skill. Table 7 gives the complete lookup table estimated from the corpus.

Table 7: Method J lookup table: $\hat{P}(Y_{nj} = 1 \mid X_{n\cdot})$ for each of the $2^3 = 8$ EP patterns and 4 skills, with the resulting $\text{EPSPI}_n^{(J)}$.

EP pattern	PS	Cr	CL	CT	$\text{EPSPI}_n^{(J)}$
$(0, 0, 0)$ None	0.204	0.015	0.041	0.245	0.126
$(0, 0, 1)$ Con only	0.628	0.054	0.536	0.390	0.402
$(0, 1, 0)$ Cog only	0.162	0.005	0.004	0.428	0.150
$(0, 1, 1)$ Cog+Con	0.932	0.113	0.599	0.855	0.625
$(1, 0, 0)$ Beh only	0.296	0.004	0.022	0.046	0.092
$(1, 0, 1)$ Beh+Con	0.693	0.003	0.892	0.059	0.412
$(1, 1, 0)$ Beh+Cog	0.340	0.011	0.012	0.162	0.131
$(1, 1, 1)$ All three	0.691	0.045	0.550	0.396	0.420

Line-level score. The Method J score for line n is the mean predicted skill probability across all $K = 4$ skills, read directly from the lookup table:

$$\text{EPSPI}_n^{(J)} = \frac{1}{K} \sum_{j=1}^{K} \hat{P}^{(J)}(Y_{nj} = 1 \mid X_{n1}, X_{n2}, X_{n3}), \tag{63}$$

where $\hat{P}^{(J)}(Y_{nj} = 1 \mid X_{n\cdot})$ is the Random Forest ensemble mean for the EP pattern of line n, given in the final column of Table 7.

Example: line with only Constructivism active. For $X_{n3} = 1$, $X_{n1} = X_{n2} = 0$ (row 2 of Table 7):

$$\text{EPSPI}_n^{(J)} = \tfrac{1}{4}(0.628 + 0.054 + 0.536 + 0.390) = 0.402.$$

Compare to Method A for the same line: $\text{EPSPI}_n^{(A)} = 0.510$. The lower Method J value reflects the ensemble's min_samples_leaf = 10 regularisation, which pulls sparse-pattern

predictions toward the training mean.

Example: line with Behaviourism only. For $X_{n1} = 1$, $X_{n2} = X_{n3} = 0$ (row 5 of Table 7):

$$\text{EPSPI}_n^{(J)} = \tfrac{1}{4}(0.296 + 0.004 + 0.022 + 0.046) = 0.092.$$

This is the lowest EPSPI of all eight patterns, reflecting that Behaviourism alone has weak skill-prediction power — consistent with the global Θ row ($\bar{\mu}_{\text{Beh}} = 0.116$) but attenuated further by regularisation.

Episode-level aggregation.

$$\text{EPSPI}_e^{(J)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \text{EPSPI}_n^{(J)}, \tag{64}$$

where n_e is the actual number of lines in episode e. Because the lookup table assigns the same score to all lines with the same EP pattern, within-episode variation in $\text{EPSPI}_n^{(J)}$ arises only from line-to-line variation in the EP profile $[X_{n1}, X_{n2}, X_{n3}]$ — identical to Method A in this respect.

Relationship to Methods A and E. Because Method J reduces to a lookup table, it is directly comparable to Method A, which also assigns scores based solely on the EP pattern of each line via the global Θ row means. The two methods differ in how the 8 lookup values are estimated:

- **Method A** computes each lookup entry as the row mean of the global Θ matrix: $\text{EPSPI}_n^{(A)} = \frac{1}{p_n} \sum_i X_{ni} \cdot \frac{1}{K} \sum_j \theta_{ij}$. For co-occurrence patterns (e.g. Beh+Cog), Method A averages the two EP row means.
- **Method J** estimates each lookup entry by ensemble averaging over 200 bootstrapped trees, with min_samples_leaf = 10 regularisation. For co-occurrence patterns, the ensemble can learn *non-additive* combinations — the Cog+Con entry (0.625) is higher than the sum of individual entries would suggest, reflecting a synergistic Cognitivism–Constructivism interaction that Method A cannot represent.

The CV r comparison between Methods A and J (A: 0.409; J: 0.394) indicates that these non-additive interactions provide negligible predictive benefit at the episode level, and the simpler Method A formula is preferred on parsimony grounds.

What "Train Pearson r" means for Method J. Method J has **32 free parameters**: the $8 \times 4 = 32$ cells of the lookup table in Table 7, estimated from the $N = 19{,}181$ training lines by ensemble averaging. The in-sample Train Pearson r reflects genuine in-sample fit

— the lookup table is estimated on the same data used for evaluation. In cross-validation, the lookup table is re-estimated from the 39 training episodes and applied to the 10 test episodes, giving CV $r = 0.394$ and CV MSE $= 0.009359$. Because the lookup table has only 32 cells and is estimated from $N = 19{,}181$ lines, overfitting risk is negligible at the cell level — the Train/CV gap for Method J is attributable to episode-level variance rather than lookup-table overfitting.

Properties. Method J has **32 free parameters** (the lookup table) and requires only the EP binary flags $[X_{n1}, X_{n2}, X_{n3}]$ at deployment — identical input to Method A. Its key advantage over Method A is the ability to capture non-additive EP co-occurrence effects through the ensemble averaging criterion. Its key limitation is that the $2^3 = 8$ pattern constraint is binding: with only 3 binary inputs, the Random Forest cannot exploit any information beyond what is already encoded in the EP pattern, making it equivalent to a lookup table regardless of ensemble size. The min_samples_leaf = 10 regularisation prevents overfitting to rare patterns (e.g. Beh+Con, which occurs in only 0.4% of lines) but also pulls predictions toward the training mean, explaining the lower predictions relative to Method A for Constructivism-active patterns. Method K (SVM) extends the predictive modelling approach by augmenting the 3 EP flags with 12 episode-conditional $\hat{\theta}_{ij}^{e_n}$ features, providing richer input at the cost of the leakage-free estimation protocol.

6.14 Method K: Support Vector Machine (SVM)

Methods E (Logistic Regression) and F (Naïve Bayes) optimise a likelihood criterion; Method J (Random Forest) optimises a tree-ensemble criterion. Method K optimises a fundamentally different objective: the **maximum-margin criterion**, which finds the hyperplane in feature space that separates skill-present from skill-absent lines with the largest possible margin [29**?**]. This geometric perspective makes SVM the most expressive discriminative classifier in the framework: it can represent non-linear decision boundaries through the kernel trick without explicitly computing high-dimensional feature maps, and its regularisation parameter C controls the bias–variance trade-off through margin width rather than through a likelihood penalty. Method K is the third and final method on the Predictive Modelling axis, completing the systematic comparison of parametric (E, F), non-parametric ensemble (J), and kernel-based (K) approaches to predicting skill presence from EP structure.

Feature construction. Method K uses the same 15-feature input as Method J (Equation (58)):

$$\mathbf{x}_n = \big[\underbrace{X_{n1},\, X_{n2},\, X_{n3}}_{\text{EP flags } (P=3)},\quad \underbrace{\hat{\theta}_{11}^{e_n},\, \hat{\theta}_{12}^{e_n},\, \ldots,\, \hat{\theta}_{PK}^{e_n}}_{\text{episode-conditional } \hat{\Theta}^{e_n} \text{ flattened } (P\times K=12)}\big] \in \mathbb{R}^{15}, \tag{65}$$

where $\hat{\theta}_{ij}^{e_n}$ is the Laplace-smoothed episode-conditional probability (Equation (**??**)). Features are standardised to zero mean and unit variance before fitting — SVM is sensitive to feature scale because the margin width is measured in Euclidean distance, and unstandardised features with different ranges would distort the margin geometry. The three EP flags ($X_{ni} \in \{0, 1\}$) and the twelve $\hat{\theta}_{ij}^{e_n}$ values (continuous in $[0, 1]$) are standardised jointly on the training fold, with the training-fold mean and standard deviation applied to the test fold without refitting.

Leakage-free episode-conditional features. The leakage-free protocol is identical to Method J. In cross-validation, for each test episode e, $\hat{\theta}_{ij}^{e}$ is computed from the 39 training episodes only, ensuring that test-episode Y_{nj} observations never enter the feature construction. Within training episodes, leave-one-out (LOO) episode-conditional estimates $\hat{\theta}_{ij}^{-e_n}$ are used: the contribution of line n to the count $\sum_{\{n':e_{n'}=e_n\}} X_{n'i}Y_{n'j}$ is removed before computing $\hat{\theta}_{ij}^{e_n}$ for that specific line. The LOO estimates are computed efficiently using the identity:

$$\hat{\theta}_{ij}^{-e_n} = \frac{\left(\sum_{\{n':e_{n'}=e_n\}} X_{n'i}Y_{n'j} + \alpha\right) - X_{ni}Y_{nj}}{\left(\sum_{\{n':e_{n'}=e_n\}} X_{n'i} + K\alpha\right) - X_{ni}}, \tag{66}$$

where $\alpha = 1$ is the Laplace smoothing constant. Equation (73) subtracts the target line's contribution from both numerator and denominator in $O(1)$ per line, given the pre-computed episode totals.

Model specification. For each skill S_j, an SVM with RBF (radial basis function) kernel is fitted:

$$f^{(j)}(\mathbf{x}_n) = \sum_{s \in \mathcal{S}^{(j)}} \alpha_s^{(j)} y_s^{(j)}\, k(\mathbf{x}_s, \mathbf{x}_n) + b^{(j)}, \tag{67}$$

where $\mathcal{S}^{(j)}$ is the set of support vectors for skill j, $\alpha_s^{(j)} \geq 0$ are the Lagrange multipliers from the dual optimisation, $y_s^{(j)} \in \{-1, +1\}$ is the binary skill label (recoded from $Y_{nj} \in \{0, 1\}$), $b^{(j)}$ is the bias term, and $k(\mathbf{x}_s, \mathbf{x}_n)$ is the RBF kernel:

$$k(\mathbf{x}_s, \mathbf{x}_n) = \exp\left(-\gamma \|\mathbf{x}_s - \mathbf{x}_n\|^2\right). \tag{68}$$

The SVM is fitted as a soft-margin classifier that allows misclassification subject to a penalty C per violation [**?**]. Hyperparameters C and γ are selected by nested cross-validation: an inner 3-fold CV on the training fold searches over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1\}$, selecting the pair minimising inner-fold MSE. This nested procedure ensures that hyperparameter selection does not use any information from the outer test fold.

Calibrated probability output. The SVM decision function $f^{(j)}(\mathbf{x}_n)$ produces a margin score, not a probability. Probability estimates are obtained via **Platt scaling** [**?**]:

$$\hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n) = \sigma\big(A^{(j)} f^{(j)}(\mathbf{x}_n) + B^{(j)}\big)\,, \tag{69}$$

where $A^{(j)}$ and $B^{(j)}$ are fitted by minimising the cross-entropy between the Platt-calibrated probabilities and the binary labels Y_{nj} on a held-out calibration set (separate from the SVM training set). Platt scaling introduces $2K = 8$ additional calibration parameters beyond the SVM decision function parameters, all estimated in a leakage-free manner within each cross-validation fold.

Line-level score. The Method K score for line n is the mean calibrated posterior skill probability across all K skills:

$$\mathrm{EPSPI}_n^{(K)} = \frac{1}{K} \sum_{j=1}^{K} \hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n), \tag{70}$$

where each $\hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n)$ is the Platt-calibrated probability from Equation (69). The formula is structurally identical to Methods E, F, and J — the mean posterior skill probability — with the difference lying entirely in how the posterior is estimated: via margin + Platt scaling for Method K, versus likelihood maximisation (E, F) or ensemble averaging (J).

Example: line with only Constructivism active in a high-Constructivism episode. For a line with $X_{n3} = 1$, $X_{n1} = X_{n2} = 0$ in Episode 13 (where $\hat{\theta}_{3j}^{e_{13}}$ is large for all j), the standardised feature vector $\mathbf{x}_n$ has a distinctive profile — high values on X_{n3} and all $\hat{\theta}_{3j}^{e_n}$ features — that lies far from the decision boundary, producing a large positive margin score $f^{(j)}(\mathbf{x}_n) \gg 0$ and a calibrated probability $\hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n) \approx 1$ for skills strongly associated with Constructivism. Such extreme predictions are exactly what inflates the in-sample Train r — the SVM memorises these high-leverage episodes through their support vectors.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(K)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(K)}, \tag{71}$$

where n_e is the actual number of lines in episode e. As with Methods E, F, I, and J, $\mathrm{EPSPI}_n^{(K)}$ varies line by line, so the episode mean is a genuine average of heterogeneous line scores.

The Train/CV gap: why Method K overfits. Method K achieves the highest Train Pearson $r = 0.853$ of all eleven methods but the lowest CV $r = 0.307$ — the largest Train/CV gap in the framework. This gap has a precise structural explanation.

The RBF kernel $k(\mathbf{x}_s, \mathbf{x}_n) = \exp(-\gamma\|\mathbf{x}_s - \mathbf{x}_n\|^2)$ measures similarity in the 15-dimensional feature space. With $E = 49$ episodes, the high-leverage episodes 13 and 49 (Section **??**) have distinctive feature profiles — extreme $\hat{\theta}_{3j}^{e_n}$ values from their dominant Constructivism — that become the support vectors around which the margin is constructed. In-sample, the SVM assigns these episodes near-perfect probability and the remaining 47 episodes moderate probabilities, producing a high Train r. Out-of-sample in CV, the test fold contains the same high-leverage episodes in approximately $10/49 \approx 20\%$ of folds; when they appear in the training fold the decision boundary is built around their extreme feature profiles, but when they appear in the test fold the boundary trained on 39 episodes predicts them as moderately Constructivist rather than maximally so — because the LOO-theta protocol prevents their own Y_{nj} values from entering the training-fold features. The $\Delta r = 0.853 - 0.307 = 0.546$ gap therefore quantifies the degree to which the SVM's in-sample performance is driven by memorising the two high-leverage episodes through their support vectors, rather than generalising the EP–skill structure to unseen episodes.

This overfitting is *not* a failure of the leakage-free protocol — the protocol correctly prevents data leakage. It reflects a fundamental limitation of kernel methods on small episode samples ($E = 49$): the RBF kernel has sufficient capacity to fit the high-leverage episodes exactly in training but insufficient regularisation (at the episode level) to generalise across folds.

Relationship to Method J. Methods J and K use identical features (Equation (72)), identical leakage-free protocols, and identical episode aggregation. Their CV r difference (J: 0.394; K: 0.307) is therefore a pure comparison of ensemble criterion vs margin criterion on the same inputs. Three factors explain why Random Forest outperforms SVM in this setting:

- **Episode imbalance.** The RBF kernel gives high similarity weight to lines in the same episode, amplifying the influence of large episodes (up to $n_e = 587$ lines) in the margin construction. Random Forest's bootstrap sampling naturally downweights the contribution of any single episode through the variance of bootstrap samples.

- **Binary feature sparsity.** With only $2^3 = 8$ distinct EP flag profiles, many training lines share identical $[X_{n1}, X_{n2}, X_{n3}]$ values and differ only in their episode-conditional $\hat{\theta}_{ij}^{e_n}$ features. Tree splits handle this mixed discrete–continuous feature space naturally; the Euclidean distance in the RBF kernel treats binary and continuous features on the same scale after standardisation, which may under- or over-weight the EP flag dimensions.

- **High-leverage memorisation.** As discussed above, the RBF kernel memorises the high-leverage episodes through dedicated support vectors; the ensemble averaging in Random Forest spreads the influence of high-leverage lines across $T = 500$ trees, reducing their individual impact on the final prediction.

What "Train Pearson r" means for Method K. Method K has no closed-form parameter count — the number of support vectors $|\mathcal{S}^{(j)}|$ varies with the data, and for $N = 19{,}181$ lines and $K = 4$ skills the total number of support vectors can range from hundreds to thousands. The in-sample Train Pearson $r = 0.853$ is the highest of all eleven methods, reflecting the SVM's capacity to fit the training data closely through the kernel trick. Unlike Methods E and I where the Train/CV gap (≈ 0.30) is attributable to episode-level overfitting of 16 parameters to 49 episodes, the SVM's gap (0.546) is attributable to kernel memorisation of high-leverage episodes — a qualitatively different form of overfitting that the nested CV hyperparameter selection does not fully prevent at the episode level. The AIC for Method K (62,315) is by far the highest of all eleven methods: the compressed $\hat{P}(Y_{nj} = 1 \mid \mathbf{x}_n) \approx \{0, 1\}$ predictions from the hard-margin region produce a poorly calibrated logistic AIC, even after Platt scaling.

Properties. Method K is a kernel-based classifier with no closed-form parameter count, $2K = 8$ Platt-calibration parameters, and two hyperparameters (C, γ) selected by nested CV. Its key advantage is theoretical: the SVM margin criterion has strong generalisation bounds through VC dimension theory [29], and the RBF kernel can represent any continuous decision boundary in the limit of large N. Its key limitation in this corpus is the combination of small episode count ($E = 49$), high-leverage episodes (13 and 49), and the RBF kernel's tendency to memorise distinctive feature profiles through support vectors — producing the largest Train/CV gap of all eleven methods. Method K therefore serves a diagnostic role: its $\Delta r = 0.546$ gap establishes an upper bound on the in-sample variance that the 15-feature representation can support, confirming that Episode-level generalisation — not line-level discrimination — is the binding constraint for all methods in this framework. Together, Methods J and K complete the Predictive Modelling axis: ensemble (J) and kernel-based (K) methods on the same 15-feature input both underperform the best parametric method (I, GEE) at the episode level, confirming that the EP–skill association structure is well-captured by the linear log-odds model and that additional model complexity does not improve episode-level predictive validity in this corpus.

6.15 Method K: Support Vector Machine (SVM)

Method J (Random Forest) uses only the 3 EP flags and reduces to a lookup table over $2^3 = 8$ patterns. Method K augments this representation by appending the 12-element flattened episode-conditional probability matrix $\hat{\theta}_{ij}^{e_n}$ to the 3 EP flags, constructing a

15-dimensional feature vector that encodes both the line's own EP profile and the local EP–skill affinity of its surrounding episode. A Support Vector Machine with a **linear kernel** and regularisation parameter $C = 1$ is fitted on this 15-dimensional input, producing a direct linear prediction of $\mathrm{EPSPI}_n^{(K)}$ [29**?**]. Method K is the third and final method on the Predictive Modelling axis.

Feature construction. The 15-dimensional feature vector for line n is:

$$\mathbf{x}_n = [\underbrace{X_{n1},\, X_{n2},\, X_{n3}}_{\text{EP flags (3)}},\ \underbrace{\hat{\theta}^{e_n}_{11},\, \hat{\theta}^{e_n}_{21},\, \hat{\theta}^{e_n}_{31},\, \hat{\theta}^{e_n}_{12},\, \hat{\theta}^{e_n}_{22},\, \hat{\theta}^{e_n}_{32},\, \hat{\theta}^{e_n}_{13},\, \hat{\theta}^{e_n}_{23},\, \hat{\theta}^{e_n}_{33},\, \hat{\theta}^{e_n}_{14},\, \hat{\theta}^{e_n}_{24},\, \hat{\theta}^{e_n}_{34}}_{\hat{\Theta}^{e_n}\ \text{column-major (12)}}] \in \mathbb{R}^{15}, \quad (72)$$

where $\hat{\theta}^{e_n}_{ij}$ is the Laplace-smoothed episode-conditional conditional probability (Equation (**??**)) for the episode containing line n. The 12 elements of $\hat{\Theta}^{e_n}$ are ordered column-major (Skill 1 across all EPs, then Skill 2, etc.), matching the implementation in Cell 23.

Leakage-free episode-conditional features. $\hat{\theta}^{e_n}_{ij}$ is estimated from the skill observations Y_{nj} of episode e_n. Using the episode's own Y_{nj} values to construct features for the same episode would create a direct within-episode dependency between predictor and outcome. Method K therefore uses the **leave-one-episode-out** (LOO) estimate $\hat{\theta}^{-e_n}_{ij}$: the conditional probability computed from all lines *except* those belonging to episode e_n:

$$\hat{\theta}^{-e_n}_{ij} = \frac{\displaystyle\sum_{\{n' : e_{n'} \neq e_n\}} X_{n'i}\, Y_{n'j} \;+\; \alpha}{\displaystyle\sum_{\{n' : e_{n'} \neq e_n\}} X_{n'i} \;+\; K\alpha}, \quad (73)$$

where $\alpha = 1$ is the Laplace smoothing constant. Equation (73) removes the entire episode e_n from the count — not just line n — so that the feature for every line in episode e_n is computed from the remaining $E - 1 = 48$ episodes. This is computed efficiently using the identity:

$$\hat{\theta}^{-e_n}_{ij} = \frac{\left(\sum_{n'=1}^{N} X_{n'i} Y_{n'j} + \alpha\right) - \sum_{\{n' : e_{n'} = e_n\}} X_{n'i} Y_{n'j}}{\left(\sum_{n'=1}^{N} X_{n'i} + K\alpha\right) - \sum_{\{n' : e_{n'} = e_n\}} X_{n'i}},$$

which subtracts the episode e_n totals from the global totals in $O(1)$ per episode, given pre-computed episode sums.

Deployment substitution. At deployment on a new episode e^* not in the training corpus, the LOO estimate $\hat{\theta}^{-e^*}_{ij}$ defaults to the published global θ_{ij} (Table **??**). This substitution is valid for prospective use: a new episode has no prior skill observations, so the global θ_{ij} is the best available estimate of the EP–skill affinity.

MinMax feature scaling. SVM with a linear kernel is sensitive to feature scale because the margin width is measured in Euclidean distance in the feature space. Each feature k is scaled to $[0, 1]$ by MinMax normalisation:

$$\tilde{x}_{nk} = \frac{x_{nk} - x_k^{\min}}{x_k^{\max} - x_k^{\min}}, \tag{74}$$

where $x_k^{\min}$ and $x_k^{\max}$ are the minimum and maximum values of feature k across all training lines. In cross-validation, scaling bounds are computed from the training fold and applied to the test fold without refitting, preventing any leakage of test-fold feature ranges into the scaling. The published scaling bounds for deployment are given in Table 8.

Linear prediction. With a linear kernel, the SVM decision function reduces to a weighted sum of scaled features plus a bias:

$$\mathrm{EPSPI}_n^{(K)} = \mathbf{w}^\top \tilde{\mathbf{x}}_n + b, \tag{75}$$

where $\mathbf{w} \in \mathbb{R}^{15}$ is the weight vector and $b \in \mathbb{R}$ is the bias, both estimated by minimising the soft-margin SVM objective with $C = 1$ [**?**]. The prediction $\mathrm{EPSPI}_n^{(K)}$ is a continuous score — not a probability — and is used directly as the episode-level predictor after aggregation. The fitted weights, scaling bounds, and bias are given in Table 8.

Table 8: Method K SVM weights w_k, MinMax scaling bounds, and feature labels. Bias $b = 0.2418$.

k	Feature	w_k	$x_k^{\min}$	$x_k^{\max}$
1	X_{n1} (Beh)	-0.018	0.000	1.000
2	X_{n2} (Cog)	$+0.033$	0.000	1.000
3	X_{n3} (Con)	$+0.499$	0.000	1.000
4	$\hat{\theta}_{11}^{e_n}$ (Beh, PS)	-0.077	0.300	0.343
5	$\hat{\theta}_{12}^{e_n}$ (Beh, Cr)	-0.063	0.005	0.008
6	$\hat{\theta}_{13}^{e_n}$ (Beh, CL)	$+0.378$	0.020	0.038
7	$\hat{\theta}_{14}^{e_n}$ (Beh, CT)	$+0.056$	0.081	0.102
8	$\hat{\theta}_{21}^{e_n}$ (Cog, PS)	-0.249	0.198	0.224
9	$\hat{\theta}_{22}^{e_n}$ (Cog, Cr)	$+0.057$	0.004	0.007
10	$\hat{\theta}_{23}^{e_n}$ (Cog, CL)	-0.077	0.002	0.010
11	$\hat{\theta}_{24}^{e_n}$ (Cog, CT)	-0.318	0.338	0.369
12	$\hat{\theta}_{31}^{e_n}$ (Con, PS)	$+0.151$	0.646	0.930
13	$\hat{\theta}_{32}^{e_n}$ (Con, Cr)	$+0.682$	0.004	0.076
14	$\hat{\theta}_{33}^{e_n}$ (Con, CL)	-0.513	0.778	0.935
15	$\hat{\theta}_{34}^{e_n}$ (Con, CT)	-0.540	0.247	0.500
Bias b		$+0.242$		

Interpreting the weight vector. Several patterns in Table 8 are scientifically interpretable:

- $w_3 = +0.499$ (Con flag) is by far the largest positive weight, confirming that a Constructivist line is the strongest single predictor of high EPSPI — consistent with the NMF latent mode (Section 6.10).
- $w_{13} = +0.682$ ($\hat{\theta}^{e_n}_{32}$: Constructivism–Creativity affinity in the local episode) is the largest positive weight among the $\hat{\theta}^{e_n}_{ij}$ features, indicating that episodes with unusually high Constructivism–Creativity co-occurrence predict high EPSPI above and beyond the line-level Con flag.
- $w_{14} = -0.513$ and $w_{15} = -0.540$ ($\hat{\theta}^{e_n}_{33}$: Con–CL and $\hat{\theta}^{e_n}_{34}$: Con–CT) are large negative weights — a counter-intuitive finding. These negative weights reflect a suppressor effect: when both $\hat{\theta}^{e_n}_{33}$ and $w_3 X_{n3}$ are large, the negative weight on $\hat{\theta}^{e_n}_{33}$ reduces the prediction to avoid double-counting the Constructivism signal already captured by the Con flag w_3.
- $w_8 = -0.249$ and $w_{11} = -0.318$ (Cog–PS and Cog–CT episode affinities) are moderately negative, indicating that high within-episode Cognitivism–skill co-occurrence is associated with lower EPSPI, possibly because Cognitivism episodes with high skill rates are rare in the corpus and the global Cognitivism baseline (θ_{2j} low) is the more reliable signal.

Deployment example. For a new line n with $X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$ (Constructivism only) in a new episode where global θ_{ij} is substituted for $\hat{\theta}^{-e_n}_{ij}$:

1. Assemble $\mathbf{x}_n = [0, 0, 1,\ \theta_{11}, \theta_{21}, \theta_{31},\ \ldots,\ \theta_{34}]$ using Table **??**.
2. Scale each element via Equation (74) using Table 8.
3. Compute $\text{EPSPI}^{(K)}_n = \mathbf{w}^\top \tilde{\mathbf{x}}_n + 0.242$.

The dominant contribution comes from $w_3 \tilde{x}_{n3} = 0.499 \times 1.0 = 0.499$ (Con flag, already scaled to 1), partially offset by the large negative weights on $\hat{\theta}^{e_n}_{33}$ and $\hat{\theta}^{e_n}_{34}$ which will be at or near their global maximum values ($\theta_{33} = 0.878$, $\theta_{34} = 0.312$) under the global substitution.

Episode-level aggregation.

$$\text{EPSPI}^{(K)}_e = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \text{EPSPI}^{(K)}_n, \tag{76}$$

where n_e is the actual number of lines in episode e. Because $\hat{\theta}^{-e_n}_{ij}$ is constant within episode e_n (all lines use the same LOO estimate), within-episode variation in $\text{EPSPI}^{(K)}_n$ arises only

from line-to-line variation in the EP flags $[X_{n1}, X_{n2}, X_{n3}]$, weighted by the corresponding w_1, w_2, w_3 terms.

The Train/CV gap: why Method K overfits. Method K achieves the highest Train Pearson $r = 0.853$ of all eleven methods but CV $r = 0.307$ — the largest Train/CV gap ($\Delta r = 0.546$). The linear SVM minimises the hinge loss on training lines, which assigns large positive margin scores to high-Constructivism lines and large negative scores to low-skill lines. The high-leverage episodes 13 and 49 (Section **??**) dominate the margin boundary: their extreme $\hat{\theta}_{ij}^{-e_n}$ feature values — far from the global θ_{ij} — become the support vectors around which $\mathbf{w}$ is fitted. In-sample, these episodes receive near-perfect EPSPI predictions, inflating Train r. In CV, when episodes 13 and 49 appear in the test fold, the boundary trained on 39 episodes produces their LOO $\hat{\theta}_{ij}^{-e_n}$ features via Equation (73) — which already differ from their in-sample features because the full episode was present during in-sample fitting but removed during LOO feature construction. The $\Delta r = 0.546$ gap quantifies the degree to which the linear SVM's in-sample performance is driven by fitting the weight vector $\mathbf{w}$ to the high-leverage episode feature profiles, which are sufficiently distinctive in the 15-dimensional space that the margin is anchored to them.

What "Train Pearson r" means for Method K. Method K has **16 free parameters**: the 15 weights w_k and bias b, estimated by the soft-margin SVM objective on all $N = 19{,}181$ training lines. The in-sample Train Pearson $r = 0.853$ reflects genuine in-sample fit under the SVM hinge-loss criterion. The AIC for Method K (62,315) is by far the highest of all eleven methods: the SVM produces predictions outside $[0, 1]$ for some lines (the linear prediction $\mathbf{w}^\top \tilde{\mathbf{x}}_n + b$ is unconstrained), which produces extreme values in the external logistic AIC model and inflates the AIC criterion.

Properties. Method K has **16 free parameters** ($\mathbf{w}$ and b) and is the only method in the framework whose EPSPI score is not constrained to $[0, 1]$: the linear SVM output $\mathbf{w}^\top \tilde{\mathbf{x}}_n + b$ can in principle fall outside this range for extreme feature vectors. Its key advantage over Method J is the richer 15-feature input: the $\hat{\theta}_{ij}^{-e_n}$ features allow the SVM to distinguish lines from different episodes even when their EP flags are identical — something Method J cannot do with its 3-flag lookup table. Its key limitation is the largest Train/CV gap of all eleven methods ($\Delta r = 0.546$), attributable to the linear margin fitting around the high-leverage episode feature profiles. The interpretable weight vector $\mathbf{w}$ (Table 8) is the primary scientific contribution of Method K beyond its predictive performance: it quantifies the relative importance of global EP structure (w_1, w_2, w_3) versus local episode context (w_4–w_{15}) in predicting skill deployment within a single interpretable linear model. Together, Methods J and K complete the Predictive Modelling axis: Method

J establishes that the 3-flag lookup table has CV $r = 0.394$, and Method K establishes that augmenting with episode-conditional features under a linear model produces CV $r = 0.307$ — confirming that the additional features do not improve episode-level generalisation and that the high-leverage episodes are the binding constraint on out-of-sample performance.

References

[1] A. Agresti. *Categorical Data Analysis.* Wiley, 2nd edition, 2002.

[2] H. Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974.

[3] R. S. J. D. Baker and K. Yacef. The state of educational data mining in 2009: A review and future visions. *Journal of Educational Data Mining*, 1(1):3–17, 2010.

[4] C. M. Bishop. *Pattern Recognition and Machine Learning.* Springer, 2006.

[5] L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[6] L. Breiman. Statistical modeling: The two cultures. *Statistical Science*, 16(3):199–231, 2001.

[7] K. P. Burnham and D. R. Anderson. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach.* Springer, 2nd edition, 2002.

[8] G. Casella and R. L. Berger. *Statistical Inference.* Duxbury, 2nd edition, 2002.

[9] T. Chai and R. R. Draxler. Root mean square error (rmse) or mean absolute error (mae)? *Geoscientific Model Development*, 7(3):1247–1250, 2014.

[10] H. Drucker, C. J. C. Burges, L. Kaufman, A. Smola, and V. Vapnik. Support vector regression machines. *Advances in Neural Information Processing Systems*, 9:155–161, 1997.

[11] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[12] M. J. Greenacre. *Theory and Applications of Correspondence Analysis.* Academic Press, 1984.

[13] D. J. Hand and K. Yu. Idiot's bayes — not so stupid after all? *International Statistical Review*, 69(3):385–398, 2001.

[14] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning.* Springer, 2nd edition, 2009.

[15] J. Hattie. *Visible Learning: A Synthesis of Over 800 Meta-Analyses Relating to Achievement*. Routledge, 2009.

[16] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant. *Applied Logistic Regression*. Wiley, 3rd edition, 2013.

[17] I. T. Jolliffe. *Principal Component Analysis*. Springer, 2nd edition, 2002.

[18] M. T. Kane. Validating the interpretations and uses of test scores. *Journal of Educational Measurement*, 50(1):1–73, 2013.

[19] S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine learning-based science. *Patterns*, 4(9), 2023.

[20] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):1–21, 2012.

[21] D. D. Lee and H. S. Seung. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401:788–791, 1999.

[22] K.-Y. Liang and S. L. Zeger. Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1):13–22, 1986.

[23] C. E. McCulloch, S. R. Searle, and J. M. Neuhaus. *Generalized, Linear, and Mixed Models*. Wiley, 2nd edition, 2008.

[24] S. Messick. Validity of psychological assessment. *American Psychologist*, 50(9):741–749, 1995.

[25] S. W. Raudenbush and A. S. Bryk. *Hierarchical Linear Models*. Sage, 2nd edition, 2002.

[26] C. Romero and S. Ventura. Educational data mining: A review of the state of the art. *IEEE Transactions on Systems, Man, and Cybernetics*, 40(6):601–618, 2010.

[27] G. Schwarz. Estimating the dimension of a model. *Annals of Statistics*, 6(2):461–464, 1978.

[28] U. Simonsohn, J. P. Simmons, and L. D. Nelson. Specification curve analysis. *Nature Human Behaviour*, 4:1208–1214, 2020.

[29] V. N. Vapnik. *The Nature of Statistical Learning Theory*. Springer, 1995.

[30] C. J. Willmott. Some comments on the evaluation of model performance. *Bulletin of the American Meteorological Society*, 63(11):1309–1313, 1982.

[31] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

[32] S. L. Zeger, K.-Y. Liang, and P. S. Albert. Models for longitudinal data: A generalized estimating equation approach. *Biometrics*, 44(4):1049–1060, 1988.